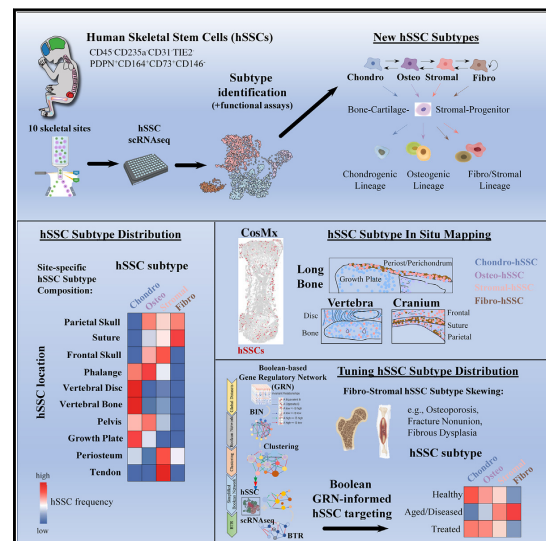


# Human skeletal development and regeneration are shaped by functional diversity of stem cells across skeletal sites

## Graphical abstract



## Highlights

- Single-cell RNA sequencing of hSSCs across ten skeletal sites and adulthood
- Identification, mapping, and functional assessment of four principal hSSC subtypes
- A fibrogenic hSSC subtype switch drives age- and disease-related skeletal impairments
- New Boolean gene regulatory network analysis informs the approach to target impaired hSSCs

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## In brief

Ambrosi and colleagues profile human skeletal stem cells (hSSCs) across ten fetal skeletal sites and from patients throughout adulthood, identifying, mapping, and functionally testing four distinct hSSC subtypes. Skeletal aging and disease are characterized by a dominant fibrogenic hSSC variant, but targeting defined gene regulatory networks reinstates functional hSSC diversity.

## Resource

# Human skeletal development and regeneration are shaped by functional diversity of stem cells across skeletal sites

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## SUMMARY

The skeleton is one of the most structurally and compositionally diverse organ systems in the human body, depending on unique cellular dynamics. Here, we integrate prospective isolation of human skeletal stem cells (hSSCs; CD45<sup>+</sup>CD235a<sup>+</sup>TIE2<sup>+</sup>CD31<sup>+</sup>CD146<sup>+</sup>PDPN<sup>+</sup>CD73<sup>+</sup>CD164<sup>+</sup>) from ten skeletal sites with functional assays and single-cell RNA sequencing (scRNA-seq) analysis to identify chondrogenic, osteogenic, stromal, and fibrogenic subtypes of hSSCs during development and their linkage to skeletal phenotypes. We map the distinct composition of hSSC subtypes across multiple skeletal sites and demonstrate their unique *in vivo* clonal dynamics. We find that age-related changes in bone formation and regeneration disorders stem from a pathological fibroblastic shift in the hSSC pool. Utilizing a Boolean algorithm, we uncover gene regulatory networks that dictate differences in the ability of hSSCs to generate specific skeletal tissues. Importantly, hSSC lineage dynamics are pharmacologically malleable, providing a new strategy to treat aberrant hSSC diversity central to aging and skeletal maladies.

## INTRODUCTION

Growth, maintenance, and regeneration of adult organ tissues require the constant replenishment of mature cell types by pools of resident stem cells. Revealing the identity and characterizing the ontogeny of these rare self-renewing, multipotent stem cells

in different tissues could unlock the underlying biology behind degenerative conditions, including aging and cancer. The utility of a stem cell-centric approach to understanding disease is best exemplified by studies on hematopoietic stem cells (HSCs), which give rise to the entire blood system and which have been revealed to play key roles in major aspects of



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hematopoiesis, from early development to age-related changes in immunity and blood-borne malignancies.<sup>1,2</sup> Recent advances in single-cell RNA sequencing (scRNA-seq) analysis further revealed previously overlooked cellular heterogeneity in HSC populations while highlighting new aspects of their regulation by non-hematopoietic cell types that serve as their niche.<sup>3–5</sup> scRNA-seq has also been applied to characterize the complexity of skeletal bone-resident niche populations in mice<sup>6–8</sup> and humans.<sup>9,10</sup> Nonetheless, our understanding of human skeletal stem and progenitor cells remains limited due to a variety of challenges, including ethical and experimental accessibility.

Although functionally intertwined with HSCs, the identity of a self-renewing, multipotent human skeletal stem cell (hSSC) has long been elusive. Several cell populations have been proposed to enrich stem and progenitor cell types that can create a hematopoiesis-supporting microenvironment but have relied on unspecific single markers or prior *in vitro* expansion for functional assessment.<sup>11,12</sup> Using a combination of eight cell surface markers, we recently described a distinct *bona fide* hSSC in the fetal growth plate capable of giving rise to bone, cartilage, and stromal tissue.<sup>13,14</sup> These stem cells can also be found in fracture calluses of regenerating adult bones and show a functional association with healing outcomes and decline during aging.<sup>15,16</sup> While HSC aging and blood diseases have been attributed to aberrant clonal expansion and lineage output,<sup>17,18</sup> which has been causally tied to skeletal stem cell (SSC) lineage dysfunction in mice,<sup>19,20</sup> the underlying mechanism of hSSC dysfunction remains to be discovered. Additionally, as multiple skeletogenic populations have been observed in long bones of mice,<sup>21–27</sup> a closer examination of human bone-forming stem cells is crucial to provide a better understanding of pathological changes in humans that are detrimental to skeletal health. In this study, we integrate scRNA-seq with functional, clonal, and spatial transcriptomic analysis of hSSCs to examine them during their development, aging, and disease states. Our findings reveal a striking diversity in hSSCs that is strongly associated with their specific anatomical locations. More importantly, we observe significant functional disparities among hSSCs obtained from patient samples, implying that alterations in stem cell diversity may be directly linked to various skeletal diseases. Specifically, we have identified an abnormal increase in fibro-stromal hSSC subtypes in cases of age-related bone loss, fracture nonunions, and Fibrous Dysplasia (FD). This skewed stem cell population suggests a possible pathological shift, contributing to the progression of these conditions.

## RESULTS

### Structural diversity in human bones is reflected by hSSC diversity

To investigate stem cell diversity in the human skeleton, we isolated hSSCs (CD45<sup>+</sup> CD235a<sup>+</sup> TIE2<sup>+</sup> CD31<sup>+</sup> CD146<sup>+</sup> PDPN<sup>+</sup> CD73<sup>+</sup> CD164<sup>+</sup>) from ten skeletal sites of the developing human fetus at the gestational age of 17–20 weeks and conducted functional and single-cell transcriptional analysis (Figures 1A, S1A, and S1B; Table S1).<sup>13,14</sup> Using established *in vitro* assays, we confirmed that hSSCs of different bones and skeletal regions displayed the potential to clonally expand and to differentiate

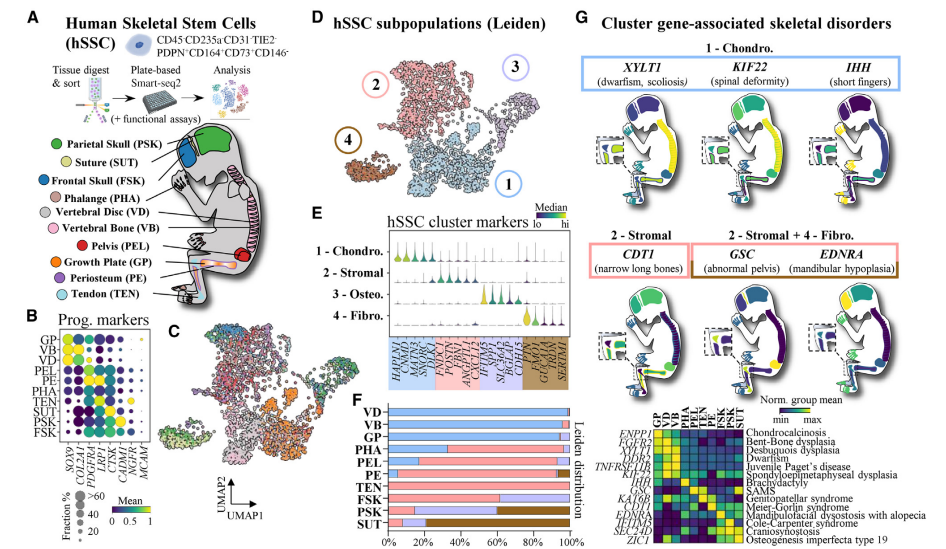
into osteogenic and chondrogenic lineages (Figures S1C–S1H). However, prospectively isolated hSSCs varied in terms of their expansion potential and their propensity to commit to either cartilage or osteoblast fates, as indicated by proteoglycan and calcium deposition, respectively.

In an effort to gain a deeper understanding of the genetic differences that underpin functional variations, we performed Smart-seq2 scRNA-seq on freshly sorted, prospectively isolated hSSCs. We discovered that hSSCs from every location investigated exhibited a minimum of 90% gene expression overlap with our previously characterized growth plate (GP) hSSCs. Remarkably, most core markers specific to GP hSSCs were also present in hSSCs from all examined skeletal sites (Figures S1I–S1K; Table S2). Additionally, our research detected the presence of the early embryonic human skeletal progenitor markers *SOX9* and *CADM1* (Figure 1B).<sup>28</sup> However, it is worth noting that there was minimal overlap with widely recognized markers to enrich for human skeletal progenitor populations.<sup>11,12,29</sup> Interestingly, two of these markers—*NGFR/CD271* and *MCAM/CD146*—were found to be significantly more concentrated in populations downstream of the hSSC (Figures 1B and S2B), including the transit-amplifying human bone-cartilage stromal progenitors (hBCSPs) and committed osteo-stromal progenitors (hOPs).

A gene expression correlation matrix as well as dimensionality reduction by uniform manifold approximation and projection (UMAP) showed distinct heterogeneity between single hSSCs from the various skeletal regions (Figures 1C and S1K). Similar to findings in mice,<sup>22,23,30,31</sup> phenotypic hSSCs from different skeletal sites expressed markers reflecting their distinct embryonic origins (Figure S1L). Unbiased Leiden clustering of all hSSCs revealed four major types representing chondrogenic, osteogenic, stromal, and fibrogenic gene expression patterns as assessed by marker expression and pathway enrichment analysis corresponding to their functional and regional differences (Figures 1D, 1E, and S1M–S1Q). The relative proportion of these hSSC subtypes closely correlated with the primary mode of ossification known for each bone, i.e., endochondral ossification (chondrogenic), which depends on a cartilage template, and direct intramembranous ossification (osteo-stromal) (Figures 1F and S1R). Thus, these results link transcriptional attributes with functional characteristics.

In attempting to associate the transcriptomic diversity of hSSCs from different locations with skeletal formation, we discovered that the expression of certain genes—which, when mutated, cause specific developmental skeletal disorders—could be detected at the stem cell level (Figures 1G and S1S). For instance, mutations in the gene *XYLT1* have been connected to dwarfism and scoliosis. This gene was found to be highly expressed in growth plate and vertebral chondral hSSC populations. In a similar fashion, the genes *CDT1* (associated with narrow long bones) and *SEC24D* (linked to craniosynostosis) demonstrated overexpression in periosteal and cranial fibro-stromal hSSCs, respectively.

Collectively, these findings underscore the notion that while hSSCs from various skeletal sites possess similar stem cell attributes, they also exhibit diversity in their genetic and lineage traits. This diversity likely contributes significantly to the regional formation, maintenance, and dysfunction of individual bones,



**Figure 1. Human bone development and disorders are rooted in distinct hSSC diversity**

(A) Schematic overview of fetal skeletal tissue compartments processed for scRNA-seq of hSSCs. (B) Dotplot showing gene expression of reported skeletal stem and progenitor (prog.) marker genes in hSSCs across skeletal sites. (C) Uniform manifold approximation and projection (UMAP) plot of 2,754 single fetal hSSCs. (D) Corresponding Leiden subclustering of all fetal hSSCs. (E) Specified Leiden subclusters with marker genes. (F) Distribution of fetal skeletal tissue-specific hSSCs in Leiden subclusters. (G) Selected differentially expressed genes associated with genetic disorders with bone phenotypes as detected in scRNA-seq data of hSSCs from different Leiden clusters. Bottom: Matrixplot showing selected genes associated with developmental skeletal disorders and their expression in site-specific hSSCs.

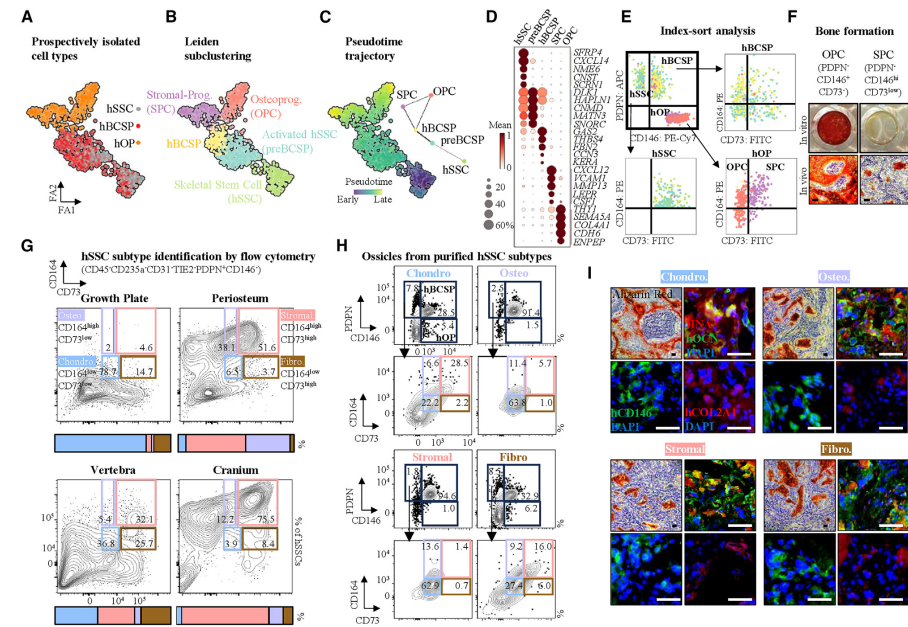
which show wide variations in shape, size, and structural composition.<sup>32</sup>

### Index-sort scRNA-seq analysis refines the hSSC lineage tree

Having observed that there are multiple subtypes of hSSCs distributed throughout the skeleton, we then asked how these variants are arranged within individual bones and how they contribute to local areas of osteogenesis. Our prior work, which examined the SSC hierarchy in the growth plate of long bones, laid out the specific differentiation pathways of mouse SSCs into chondrogenic, osteogenic, and fibro-stromal progenitors (Figure S2A).<sup>13,22</sup> To better elucidate the bone-forming lineage of hSSCs, we employed index-sorting with plate-based Smart-seq2 scRNA-seq on prospectively isolated GP hSSCs and their defined progenitor subtypes (hBCSPs and hOPs). Using UMAP clustering, Leiden separation, trajectory inference, and CytoTRACE projections, we identified hierarchical lineage relationships between hSSC and hSSC-derived progenitor cell types (Figures 2A–2D, S2B, and S2C).<sup>22</sup> Utilizing index-sort information, we further identified new cell surface marker combinations that allowed us to resolve specific hSSC-derived osteogenic

progenitor cell (OPC: PDPN<sup>+</sup> CD146<sup>+</sup> CD73<sup>+</sup>) and stromal progenitor cell (SPC: PDPN<sup>+</sup> CD146<sup>hi</sup> CD73<sup>low</sup>) subsets from the hOP population (PDPN<sup>+</sup> CD146<sup>+</sup>) (Figures 2E, S2D, and S2E). In support of this finding, prospectively isolated OPCs were highly osteogenic compared with SPCs *in vitro* and *in vivo* (Figures 2F, S2F, and S2G). Furthermore, utilizing our scRNA-seq-based lineage reconstruction together with our previously published single-cell data of micro-dissected sections of the growth plate zones and the adjacent diaphysis,<sup>13</sup> we were able to spatially map the lineage hierarchy from fetal hSSCs to preBCSPs, BCSPPs, and their committed OPC and SPC downstream subsets. Thus, GP hSSCs exhibit a unique sequential differentiation pattern stretching from the epiphysis to the diaphysis. This differentiation process generates a clearly organized stratification of hSSC-derived progenitors within the maturing growth plate (Figures S2H–S2K).

Since scRNA-seq data inferred the existence of four hSSC subtypes (Figures 1D and 1E), we wondered if they could also be identified based on our existing flow cytometric panel for prospective isolation and functional characterization. Since index-sort analysis of the cartilaginous fetal growth plate skeletal lineage suggested mostly moderate CD73 and CD164

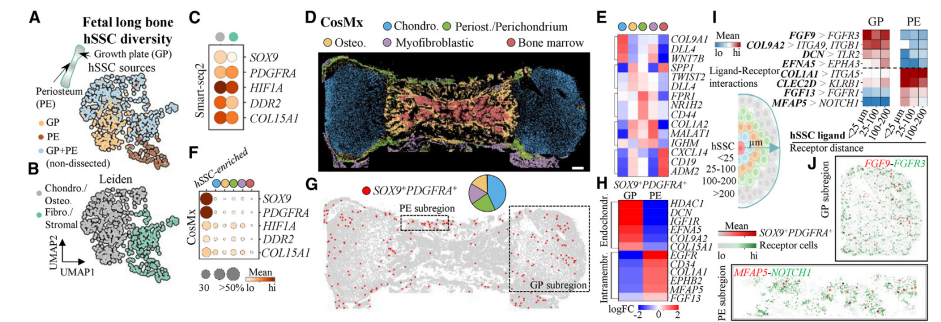


**Figure 2. The refined hSSC lineage tree contains four distinct hSSC subtypes**

(A) ForceAtlas2 projection of scRNA-seq data from previously defined and prospectively isolated human skeletal lineage populations (885 cells). hSSC, human skeletal stem cell (175 cells); hBCSP, human bone-cartilage-stromal progenitor cell (369 cells); hOP, human osteoprogenitor (341 cells). (B) Unbiased Leiden clustering of the same cells with annotation of putative cell types. (C) Pseudotime trajectory inference of scRNA-seq data from prospectively isolated hSSC lineage populations. (D) Leiden cluster-specific marker expression. (E) Index-sort analysis of scRNA-seq data from skeletal lineage populations colored by Leiden subclusters. (F) *In vitro* (top) and *in vivo* (bottom) osteogenesis of freshly purified sorted OPCs (PDPN<sup>+</sup>CD146<sup>hi</sup>CD73<sup>low</sup>), and SPCs (PDPN<sup>+</sup>CD146<sup>hi</sup>CD73<sup>low</sup>) stained with alizarin red S. (G) Representative flow cytometric plots of fetal tissue from four skeletal sites showing distinct hSSC subtype distribution within phenotypic hSSC gate (CD45<sup>+</sup>CD235a<sup>+</sup>CD31<sup>+</sup>TIE2<sup>+</sup>PDPN<sup>+</sup>CD146<sup>+</sup>CD73<sup>+</sup>CD164<sup>+</sup>). Below: composition (%) of hSSC subtypes within phenotypic hSSC gate. (H) Flow cytometric analysis of 4-week ossicles derived from the four distinct hSSC subtypes. (I) Representative alizarin red S staining and immunohistological staining supporting osteogenic (hOCN), chondrogenic (hCOL2A1), and stromal (hCD146) output (HNA, human nuclear antigen; hOCN, osteocalcin; hCOL2A1, collagen 2A1; DAPI, nuclei) of sections from 8-week *in vivo* ossicles derived from prospectively isolated hSSC subtype populations. Scale bars, 50  $\mu$ m.

expression levels in hSSCs (Figure 2E), we further subdivided the double-positive hSSC fraction from pooled fetal long bones, vertebrae, and cranium into four subpopulations with varying CD73/CD164 expression intensity and prospectively isolated them for quantitative PCR analysis. We found that scRNA-seq gene expression matched CD164<sup>low</sup>CD73<sup>low</sup> (chondrogenic), CD164<sup>high</sup>CD73<sup>low</sup> (osteogenic), CD164<sup>high</sup>CD73<sup>high</sup> (stromal), and CD164<sup>low</sup>CD73<sup>high</sup> (fibrogenic) hSSC populations (Figures S2L and S2M). Based on these cell surface marker profiles, and as suggested by scRNA-seq, flow cytometric analysis demonstrated that growth plate and vertebral hSSCs mainly contained the chondrogenic subtype while periosteum (PE) and cranial hSSCs were highly enriched

for stromal hSSCs (Figure 2G). To test the plasticity of the hSSC subtype composition, we challenged hSSCs from different skeletal source tissues with a pro-osteogenic differentiation cocktail, which was sufficient to generate large proportions of the osteogenic hSSC subtype independent of original variant distribution at the four skeletal sites (Figure S2N). We independently confirmed *bona fide* SSC properties for each hSSC subtype by prospectively isolating them and performing *in vivo* ossicle formation assays. Both flow cytometric and histological analyses confirmed the self-renewal and multilineage differentiation capacity of all four hSSCs subtypes in this setting (Figures 2H and 2I). Altogether, these results confirm scRNA-seq data, tying distinct but



**Figure 3. Single-cell spatial mapping of fetal long bone hSSCs**

(A) UMAP of scRNA-seq data of 616 prospectively isolated growth plate (GP; 172 cells) hSSCs, periosteum (PE; 78 cells) hSSCs, and hSSCs from non-dissected long bone tissue (GP + PE; 366 cells). (B) Unbiased Leiden clustering of the same dataset. (C) Selected gene expression of hSSC-enriched markers in Leiden populations. (D) Nanostring CosMx spatial transcriptomic analysis of a fetal phalange bone section with single cells clustered using CellCharter. (E) Marker genes of the subpopulations identified by CellCharter. (F) Expression of genes enriched in hSSCs in different clusters. (G) Human SSC-enriched population detected by co-expression of SOX9 and PDGFRA shown as distribution across fetal bone. Fraction of hSSCs per cell cluster shown in pie chart. Specific subregions enriched for SOX9<sup>+</sup>PDGFRA<sup>+</sup> cells are demarcated. (H) Distinct gene expression of SOX9<sup>+</sup>PDGFRA<sup>+</sup> cells in GP and PE subregions. (I) CellPhoneDB ligand-receptor interactions inference between SOX9<sup>+</sup>PDGFRA<sup>+</sup> and surrounding cells. (J) Expression of selected ligands in SOX9<sup>+</sup>PDGFRA<sup>+</sup> cells in subregions as well as expression of respective receptors in surrounding cells. Scale bar, 1 mm.

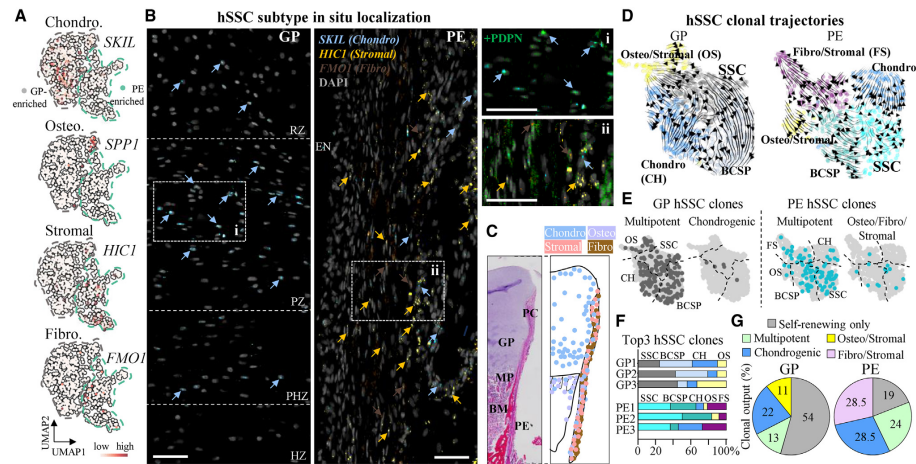
malleable hSSC subtype distribution to various skeletal sites (Figure S2O).

### Spatially mapped region-specific hSSC diversity is controlled by distinct clonal dynamics

To spatially resolve hSSC subtype distribution, we first investigated the distinct differences in GP and PE hSSC diversity (Figure S3). We integrated scRNA-seq data of hSSCs from dissected growth plate and periosteal regions as well as non-dissected long bones, which contained both anatomical sublocations. Gene expression by unbiased clustering again showed that PE hSSCs predominantly consist of fibrous and stromal subtypes, while GP hSSCs are mainly chondrogenic (Figures 3A–3C and S3A–S3F). We then conducted spatial transcriptomic analysis of a fetal phalange using Nanostring's CosMx platform with a universal 1,000 human genes panel, which together with the CellCharter algorithm<sup>35</sup> successfully mapped individual cells to specific subregions (Figures 3D, 3E, S3L, and S3M). Since we observed that the two long bone hSSC subtypes share expression of common markers (Figures 3C and S1I), we estimated hSSC locations and distribution throughout the long bone by identifying SOX9<sup>+</sup>PDGFRA<sup>+</sup> cells in the CosMx dataset. Indeed, the majority of SOX9<sup>+</sup>PDGFRA<sup>+</sup> cells could be found in growth plate and PE regions, while they were not present in the bone marrow (Figures 3F, 3G, and S3N). In accordance with Smart-seq2 results of hSSC subtypes, SOX9<sup>+</sup>PDGFRA<sup>+</sup> cells in the growth plate expressed markers associated with endochondral ossification, while cells labeled in the periosteum were enriched for genes related to intramembranous ossification (Figure 3H). This was also consistent

with the inference of specific ligand-receptor interactions with other cell types in the vicinity of labeled hSSCs (Figures 3I, 3J, and S3O) as well as potential longer-distance crosstalk between hSSC subtypes (Figures S3G–S3K). Altogether, these results estimate the spatial distribution of long bone hSSCs *in situ* and confirm distinct molecular differences of identified regional subtypes.

We then mined our Smart-seq2 datasets to identify distinct markers that could be used to approximate the anatomical organization of the newly identified hSSC subtypes across different skeletal sites. Confirming CosMx results, RNA *in situ* hybridization for SKI like proto-oncogene (*SKIL*) with co-staining of hSSC marker podoplanin (PDPN) localized the chondrogenic hSSC subtype primarily to the resting and proliferative zones of the growth plate and verified the presence of HIC ZBTB transcriptional repressor 1 (*HIC1*)-expressing stromal hSSCs along the periosteum (Figures 4A–4C, S1J, S4A, and S4B). When mapping out the *in situ* distribution of all four hSSC subtypes (i.e., chondrogenic [*SKIL*\*PDPN<sup>+</sup>], osteogenic [secreted phosphoprotein 1, *SPP1*\*PDPN<sup>+</sup>], stromal [*HIC1*\*PDPN<sup>+</sup>], and fibrogenic [Flavin containing monooxygenase 1, *FMO1*\*PDPN<sup>+</sup>]) across long bones, vertebrae, and the cranium, we could validate their overall prevalence as predicted by scRNA-seq (Figures S4A–S4G). While we found chondrogenic and osteogenic hSSC subtypes in the metaphysis close to newly forming trabecular bone, as well as sparsely across periosteal tissue, none of the four subtypes were present in the diaphyseal bone marrow space, in close alignment with CosMx data. Our analysis therefore suggests that cells previously classified as (bone marrow)



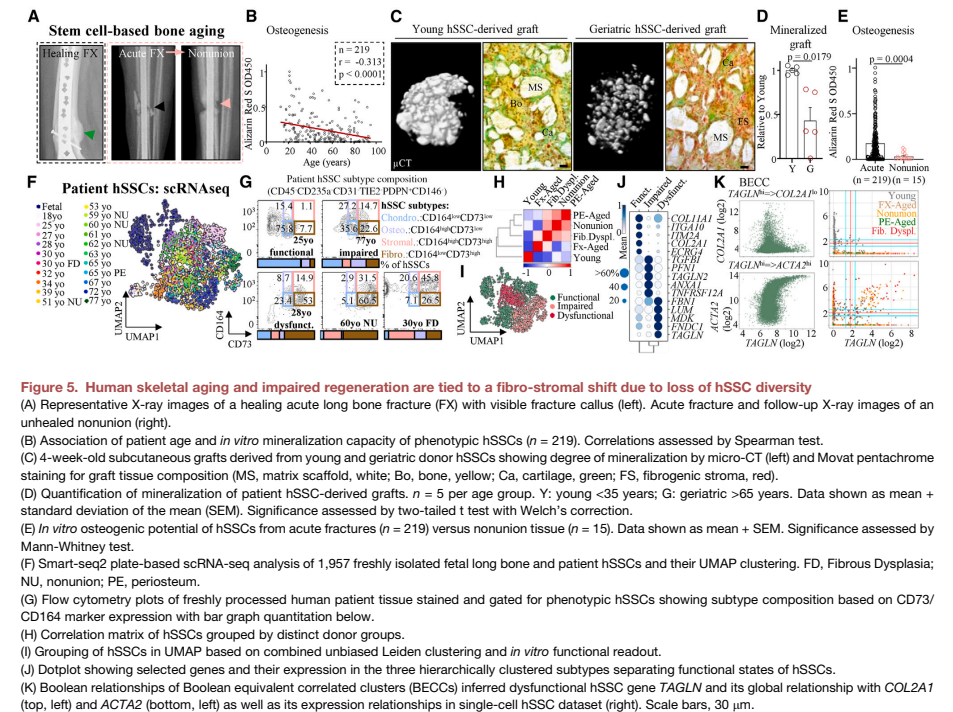
**Figure 4. Regional-specific hSSC diversity is reflected in distinct clonal dynamics**

(A) Gene expression of hSSC subtype-specific markers *SKIL* (Chondro.), *SPP1* (Osteo.), *HIC1* (Stromal), and *FMO1* (Fibro.) in UMAP of scRNA-seq data from GP and PE hSSCs. (B) *In situ* validation of hSSC subtype marker expression in fetal long bone growth plate (left) and periosteum (right) using RNAscope. DAPI, nuclear stain; RZ, resting zone; PZ, proliferative zone; PHZ, pre-hypertrophic zone; HZ, hypertrophic zone; EN, endosteal site. Insets show co-staining with hSSC marker podoplanin (PDPN). (C) H&E staining of fetal long bone (left) and RNA *in situ* hybridization-based distribution of overall prevalence of hSSC subtypes (bottom). GP, growth plate; MP, metaphysis; PE, periosteum; PC, perichondrium. (D) UMAP with RNA velocity of Leiden clustered scRNA-seq data from barcoded GP-hSSC-derived grafts (left) and barcoded PE-hSSC-derived grafts (right). SSC, skeletal stem cell; BCSP, bone-cartilage-stromal progenitor; CH, chondrogenic; OS, osteo-stromal; FS, fibro-stromal. *n* = 3 grafts per group. (E) Distribution of single-cell-derived cells within UMAP Leiden clusters from two selected GP-hSSC clones (left) and for two selected PE-hSSC clones (right). (F) The top 3 most abundant clones per hSSC site of origin and their relative cell type contributions based on Leiden clusters. (G) Overall detected clonal cell type output of barcoded GP (*n* = 46) and PE (*n* = 21) hSSCs. Scale bars, 50 µm.

mesenchymal stromal cells (BMSCs) are cell types distinct from hSSCs. BMSCs likely consist of perivascular cells (e.g., TIE2<sup>+</sup>) that could include lineage cells derived from hSSCs. The mostly chondrogenic vertebral hSSCs were highly abundant specifically at the growth plate area of vertebral bodies and concentric to the nucleus pulposus, spanning most of the annulus fibrosus of the vertebral disc (Figures S4C and S4D). By contrast, stromal and osteogenic hSSC subtypes were abundant in frontal and parietal skull regions of the cranium (Figures S4E and S4F). *FMO1*-expressing fibrogenic hSSCs were highly prevalent throughout cranial suture tissue but less in periosteal and parietal skull regions, while they were almost undetectable in growth plate, vertebra, and frontal skull regions (Figure S4G). Combined, these results connect transcriptomic, functional, and histological readouts of hSSC subtype diversity.

To study the differentiation dynamics of individual hSSCs from the growth plate and periosteum, we developed a clonal barcoding approach. This strategy allows us to trace the differentiation paths of individual genetically barcoded hSSCs *in vivo* (see STAR Methods). The single-cell readouts confirmed distinctive differentiation outcomes between GP and PE hSSCs. As ex-

pected from our hSSC subtype analysis, both hSSC populations demonstrated robust self-renewal capacity yet showed significant differences in their formation of cartilage, bone, and fibro-stromal progenitors on the single-cell level (Figures 4D–4G and S4H–S4M). For instance, growth plate-associated hSSCs mainly contribute to the development of both cartilage and osteo-stromal tissues, playing a critical role in facilitating the longitudinal growth of bones. By contrast, hSSCs found in the periosteum predominantly generate fibro-stromal lineages, providing toughness and flexibility to the cortical bones. Thus, the specific anatomical locations of hSSCs are crucial in maintaining the overall structure of bone. Importantly, when either of these specific hSSC harboring areas are eliminated through microsurgery, the overall development of long bones is significantly impaired (Figures S4N and S4O). This demonstrates the vital role these specialized hSSCs play for bone structure and growth. Altogether, these results indicate that just as different bones contain varying ratios of hSSC variants, different subtypes of hSSCs are distributed in unique domains within a long bone. Therefore, these studies underline the crucial role of location in the function of different hSSC subtypes within human bone tissue.



**Figure 5. Human skeletal aging and impaired regeneration are tied to a fibro-stromal shift due to loss of hSSC diversity**

(A) Representative X-ray images of a healing acute long bone fracture (FX) with visible fracture callus (left). Acute fracture and follow-up X-ray images of an unhealed nonunion (right). (B) Association of patient age and *in vitro* mineralization capacity of phenotypic hSSCs (*n* = 219). Correlations assessed by Spearman test. (C) 4-week-old subcutaneous grafts derived from young and geriatric donor hSSCs showing degree of mineralization by micro-CT (left) and Movat pentachrome staining for graft tissue composition (MS, matrix scaffold, white; Bo, bone, yellow; Ca, cartilage, green; FS, fibrogenic stroma, red). (D) Quantification of mineralization of patient hSSC-derived grafts. *n* = 5 per age group. Y: young <35 years; G: geriatric >65 years. Data shown as mean + standard deviation of the mean (SEM). Significance assessed by two-tailed t test with Welch's correction. (E) *In vitro* osteogenic potential of hSSCs from acute fractures (*n* = 219) versus nonunion tissue (*n* = 15). Data shown as mean + SEM. Significance assessed by Mann-Whitney test. (F) Smart-seq2 plate-based scRNA-seq analysis of 1,957 freshly isolated fetal long bone and patient hSSCs and their UMAP clustering. FD, Fibrous Dysplasia; NU, nonunion; PE, periosteum. (G) Flow cytometry plots of freshly processed human patient tissue stained and gated for phenotypic hSSCs showing subtype composition based on CD73/CD164 marker expression with bar graph quantitation below. (H) Correlation matrix of hSSCs grouped by distinct donor groups. (I) Grouping of hSSCs in UMAP based on combined unbiased Leiden clustering and *in vitro* functional readout. (J) Dotplot showing selected genes and their expression in the three hierarchically clustered subtypes separating functional states of hSSCs. (K) Boolean relationships of Boolean equivalent correlated clusters (BECCs) inferred dysfunctional hSSC gene *TAGLN* and its global relationship with *COL2A1* (top, left) and *ACTA2* (bottom, left) as well as its expression relationships in single-cell hSSC dataset (right). Scale bars, 30 µm.

### Skeletal aging and loss of regenerative activity are tied to changes in hSSC diversity

Previous studies have linked aging with decreased skeletal regeneration.<sup>15,19,34</sup> Our in-depth investigation of age-related changes in hSSCs found that, throughout adulthood, these cells retain the highest potential for osteochondrogenic development *in vitro* compared with other cell types (Figure S5A). However, an analysis of 407 patient-derived fracture callus specimens demonstrated a significant decrease in hSSC frequency with advanced age (Figure S5B). Functional *in vitro* and *in vivo* assays conducted on hSSCs isolated from 270 of these patients revealed diminished clonogenicity and bone/cartilage-forming potential in older patients (Figures 5A, 5B, and S5C–S5E). Subcutaneous transplants of hSSCs from geriatric donors into young mice yielded ossicles that showed significantly reduced mineralization compared with hSSCs from young patients (Figures 5C and 5D). This was in line with a sex-independent age-related pattern of reduced hSSC frequency and expansion potential, with male donor cells showing a significant negative correlation in colony-forming unit-fibroblast (CFU-F) capacity with age. The negative correlation of increasing age and bone formation potential was also independent of the patient's sex and body mass index (BMI) (Figures S5F and S5G).

Using functional readouts and scRNA-seq, we found that poor fracture healing, which predominantly occurs in older patients and can result in nonunions, was associated with the formation of fibrogenic tissue instead of mineralized fracture callus (Figures 5E and S5H–S5J).<sup>35</sup> Surprisingly, hSSC numbers were increased in nonunion fracture tissue, but these cells exhibited deficient osteogenesis, indicating a link between impaired hSSC differentiation and compromised bone regeneration. The results demonstrate that hSSCs from older patients primarily generate fibro-stromal tissue, suggesting that dysfunction in hSSCs could underlie pathologically shifted fibrogenic lineage dynamics, thereby impacting the major stem cell source for fracture repair.

To gain a better understanding of the biological and molecular differences in young, aged, and nonunion adult hSSCs, we conducted Smart-seq2 scRNA-seq analysis of freshly purified long bone fracture callus site-derived hSSCs from 20 patients (Table S1) along with evaluation of their osteogenic potential (Figures 5F and S5K). Based on our functional readouts, we hypothesized that higher age favored fibro-stromal differentiation of fracture hSSCs, thereby preventing proper mineralization at the healing site. This was reminiscent of the phenotype found in bones of patients with FD, a genetic disease caused by an

activating Gs alpha mutation resulting in softened bones and generation of fibrotic marrow that has been linked to stem cell dysfunction.<sup>36,37</sup> Thus, we also isolated phenotypic hSSCs from a patient with FD. We found a high prevalence of phenotypic hSSCs in FD bone. However, like nonunion hSSCs, FD hSSCs failed to undergo osteogenesis and displayed strong expression of fibroblastic genes and myofibroblastic proteins such as smooth muscle alpha-actin ( $\alpha$ SMA/ACTA2) (Figures S5L and S5M). Strikingly, functional readouts of patient hSSCs were reflected in distinct hSSC subtype composition detected in patient samples (Figure 5G), i.e., higher prevalence of the osteochondrogenic hSSC subtypes in bone-forming hSSCs compared with fibro-stromal subtype abundance in dysfunctional skeletal tissues. In support of that, the hSSCs from donors with pro-fibrogenic features, i.e., nonunion, FD, and aged PE, showed strong transcriptomic correlation (Figure 5H). Additionally, young osteogenic hSSCs (fetal and patients <35 years) showed higher developmental potential, as predicted by CytoTRACE, and lower expression of myofibroblastic genes compared with aged, nonunion, aged PE, and FD hSSCs (Figures S5N–S5P). Intriguingly, as we have previously shown that osteogenic differentiation performance of hSSCs can be a predictor of fracture union even in young patients,<sup>10</sup> we found that *in vitro* mineralization capacity assessed by alizarin red staining and fibrogenic gene marker expression strongly correlated with hierarchical clustering of single hSSCs (Figure S5Q and S5R). Indeed, hSSCs from a 28-year-old patient failed to mineralize in culture and clustered closely with pro-fibrogenic nonunion and FD hSSCs, similar to PE hSSCs from a 65-year-old patient. To investigate the defining properties of different hSSC types more easily, we grouped cells into functional (osteochondrogenic), impaired, and dysfunctional (pro-fibrogenic) hSSCs based on unbiased Leiden separation and functional readouts (Figures 5I and S5S). While stem cell marker expression was mostly conserved throughout functional groups, top regulated genes were shifted from skeletal system development in functional hSSCs to positive regulation of fibroblast differentiation and regulation of smooth muscle cell proliferation in impaired and dysfunctional hSSCs (Figures 5J, S5S, and S5T). Taken together, these results support the notion that pro-fibroblastic shifts in hSSC diversity are a major pathological mechanism of stem cell dysfunction in bones, which might create an imbalance in skeletal homeostasis resulting in impaired regenerative capacity.

Finally, since our observations revealed a shift toward fibroblast-skewed hSSCs in aging, nonunions, and FD, it prompted the question of whether this shift is a common mechanism in skeletal disorders. The fibroblast marker  $\alpha$ SMA/ACTA2 was found to be overexpressed in FD and nonunion hSSCs. To identify a universal biomarker that could potentially diagnose an impaired skeletal tissue state, we employed computational Boolean equivalent correlated clusters (BECs) using a database of nearly 26,000 public gene expression datasets.<sup>38,39</sup> This approach identified 25 genes, including the well-known fibroblast markers transgelin (*TAGLN*) and myosin light-chain 9 (*MYL9*), as potential markers of fibroblast-skewed hSSCs due to their high correlation with each other and *ACTA2*—the seed gene (Figure S5U).<sup>40</sup> Single-cell transcriptomic data confirmed the specific expression of these markers in fibro-stromal fetal hSSC subtypes as well as in impaired aged, FD, and nonunion hSSCs (Figures S5P, S5V, and S5W).

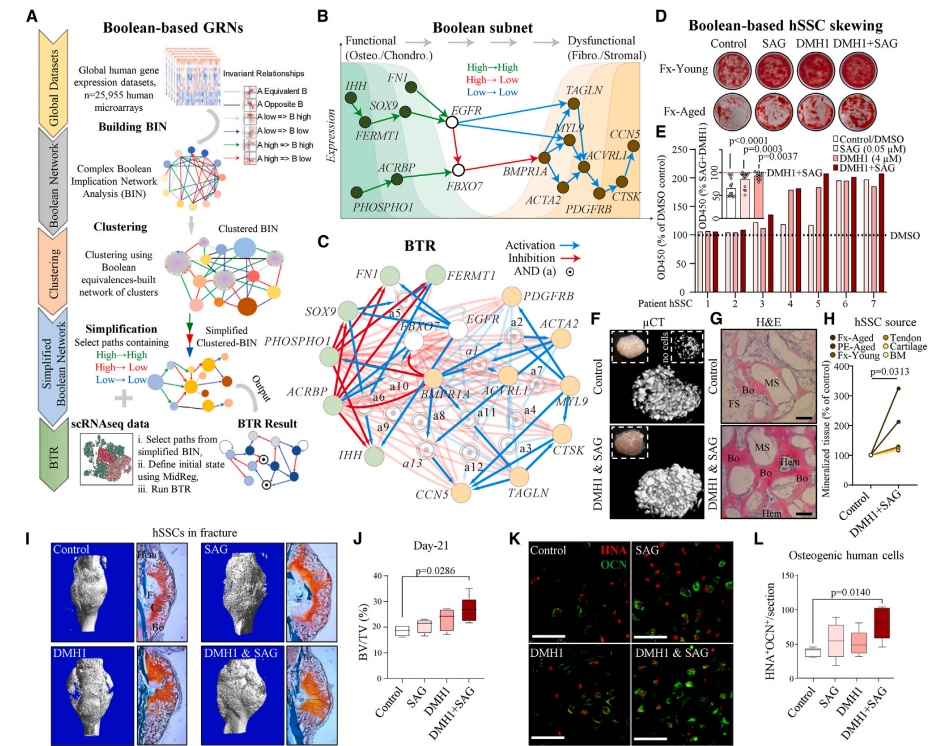
Remarkably, *TAGLN*, a pro-fibrotic gene, showed a high-to-low correlation with osteochondrogenic marker genes (e.g., *COL2A1* and *SOX5*), emphasizing the inverse Boolean relationship between these two gene programs (Figure 5K, S5X, and S5Y). Analysis of publicly available human gene expression datasets revealed low expression of *TAGLN* and *MYL9* in fetal hSSCs, growth plate zone cells, and freshly isolated bone marrow stromal cells from healthy adults.<sup>13</sup> However, these markers were highly abundant in tissue samples from various fibro- and osteosarcomas.<sup>41</sup> Notably, *TAGLN* was a better biomarker than *MYL9* for osteoporotic and osteoarthritis bone tissue (Figure S5Z). In sum, our findings suggest an expansion of fibroblast-skewed hSSCs in multiple skeletal pathologies. Thus, shifts toward higher proportions of this hSSC subtype may serve as an indicator of poor bone health.

### Combinatorial targeting of gene regulatory networks to preserve hSSC diversity

Noticing considerable transcriptomic differences between functional and dysfunctional hSSCs, we sought to identify the gene regulatory mechanisms that influence their identities. We used an innovative approach, an invariant analysis of Boolean implication relationships, across 25,955 human microarrays from various tissues and developed a new workflow integrating scRNA-seq data from healthy/osteochondrogenic and dysfunctional/fibrogenic hSSC subtypes (Figures S6A–S6I; STAR Methods for a detailed description).

Initially, we created a comprehensive Boolean implication network (BIN) using the most differentially expressed and common skeletal markers in hSSCs, revealing fundamental global gene expression relationships (Figures 6A and S6J). To simplify this, we identified BIN subnets (pathways) and employed high-level mathematical modeling to predict the gene regulation orders that linked pro-fibrogenic genes in dysfunctional hSSCs to genes expressed in functional hSSCs. Intriguingly, our results indicated that genes associated with the hedgehog (Hh) pathway (*IHH*, *PHOSPHO1*) had distinct Boolean relationships with fibrogenic genes (*ACTA2* and *TAGLN*), which were closely connected to bone morphogenic protein (BMP) signaling (*BMPRIA* and *ACVRL1*) (Figure 6B). Consistent with the vital role of Hh signaling in limb development and its deficiency as a factor in diabetic bone disease, functional hSSCs exhibited higher expression of genes within this pathway compared with their dysfunctional fibrogenic counterparts (Figure S6K). Conversely, excess BMP and heightened transforming growth factor beta (TGF- $\beta$ ) signaling are linked with fibrosis and bone loss.<sup>44,45</sup>

We then applied the BoolTrainE (BTR) models, based on Smart-seq2 scRNA-seq data, to the selected subnets to identify dynamic Boolean circuits on a single-cell level (Figure 6C).<sup>46</sup> The results highlighted an unforeseen complexity in gene regulation. They suggested the necessity of an activating *SOX9-FERMT1-IHH* gene regulatory loop in osteogenic hSSCs while simultaneously preventing suppression by excessive inhibitory *BMPRIA*-related gene programs. This was achieved through the tight control of *PHOSPHO1* expression to favor osteochondrogenic over fibrogenic cell states. To test this functionally, we used the small molecules SAG21k, which stimulates Hh



**Figure 6. Boolean-logic-guided combinatorial targeting of gene regulatory networks restores hSSC diversity**

(A) Schematic of workflow for invariant analysis of single-cell data from functional and dysfunctional patient hSSCs based on Boolean implication relationships. (B) Selected BIN pathways and high-level mathematical model of gene regulation in hSSCs with predicted orders of activation. Gene relationships indicated by color of arrows. Green, hi-to; red, hi-to; blue, lo-to. (C) BoolTrainE (BTR) model of a gene regulatory network of hSSCs based on a selected pathway identified by BIN to capture dynamic Boolean circuit-based model. (D) Representative alizarin red S staining of *in vitro* osteogenic assays on patient-derived fracture (Fx) hSSCs either treated with DMSO as control or small molecules dorsomorphin homolog 1 (DMH1), smoothened agonist 21k (SAG), and a combination of DMH1 and SAG. (E) Quantification of *in vitro* osteogenic assays with small molecule treatments on hSSCs from seven patients. Insert shows quantification of combined patient hSSCs relative to combinatorial DMH1 + SAG treatment. Data shown as mean + standard deviation of the mean (SEM). Two-tailed Student's t test versus combinatorial treatment with Welch's correction or Mann-Whitney test to correct for unequal variances or non-normality.  $n = 14$  from independent duplicates of seven patient hSSCs. (F) Representative  $\mu$ CT images of 8-week grafts derived from transplanted patient hSSCs either treated with DMSO (control) or DMH1 + SAG. Top left insert shows bright-field image of grafts. No hSSC graft shown in the top right insert of control. (G) Representative H&E staining of graft sections. MS, matrix scaffold; Bo, bone; FS, fibro/stroma; Hem, hematopoietic. (H) Quantification of hSSC grafts either treated with DMSO or combinatorial cocktail. Two-tailed paired Wilcoxon test ( $n = 6$ ). (I) Representative micro-CT and safranin-O staining images of fracture calluses with hSSC transplants with or without factors at day 21 after bi-cortical fracture. Ca, cartilage. (J) Micro-CT-based analysis of fractional bone volume in formed callus. (K) Immunohistochemistry identifying hSSC-derived cells stained for human nuclear antigen (HNA, red) and osteocalcin (OCN, green) in fracture callus tissue of different experimental groups. (L) Quantification of HNA and OCN double-positive cells across calluses. All fracture callus analyses  $n = 4$ –5 per group. Data shown as box-plots with min to max. One-way ANOVA with Dunnett's post hoc test. Scale bars, 50  $\mu$ m.

through smoothened activation, and dorsomorphin homolog 1 (DMH1), which selectively blocks BMP signaling to prevent the upregulation of fibrogenic genes. Based on our Boolean analysis, we hypothesized that a combined treatment with these molecules could restore hSSC bone-forming hSSC subtype composition in dysfunctional stem cells. *In vitro* testing confirmed that the combined application of SAG21k and DMH1 to primary patient hSSCs elevated *IHH* and *SOX9* expression while downregulating *BMPRI1A* and *TAGLN* (Figure S6L). Importantly, these small molecules significantly enhanced the osteogenesis of patient hSSCs if applied together (Figures 6D and 6E). For *in vivo* testing, we first transplanted patient-derived hSSCs encapsulated in Matrigel with acellular bone matrix, alongside SAG21k and DMH1, subcutaneously into immunodeficient NOD scid gamma (NSG) mice (Figure S6M). The combined treatment significantly improved ossicle formation with higher mineralized content compared with controls. While fracture hSSCs responded most efficiently, this dual regimen also enhanced bone formation in hSSCs isolated from tendon, cartilage, and metaphyseal bone marrow of aged donors (Figures 6F–6H). Finally, we transplanted gels encapsulating hSSCs with or without SAG21k/DMH1 around the injury site of NSG mice with a bi-cortical femoral fracture to determine cellular output of human cells and healing outcome (Figure S6N). Flow cytometric analysis of day-10 callus tissue identified donor hSSCs enriched for the chondrogenic subtype contributing to multiple skeletal lineages during the endochondral ossification process (Figure S6O). Micro-computed tomography (CT) and immunohistology of day-14 callus tissue further demonstrated the osteochondrogenic contributions of donor hSSCs during healing and importantly significantly increased mineralization through enhanced bone and cartilage output from hSSCs stimulated by SAG21k+DMH1 but not mono-treated groups compared with hSSC-only controls (Figures 6I–6L and S6P–S6R). In summary, we have pioneered a novel approach to gene regulatory network analysis based on Boolean logic, inferred from scRNA-seq data of hSSCs. This surpasses traditional methods that only infer transcription factor interactions (Figure S6S). Utilizing this advanced technique, we identified specific small molecules that can reinstate functional diversity of non-osteochondrogenic, dysfunctional hSSCs.

## DISCUSSION

Our study provides a unique view into human skeletal development, aging, and disease from the perspective of highly pure hSSCs. We have found that individual hSSCs vary in their capacity for expansion, self-renewal, and differentiation into bone, cartilage, and fibro-stromal cells but do not generate fat cells. Our research shows that while hSSCs derived from different skeletal sites display high transcriptomic similarity, each bone hosts a unique diversity of hSSC subtypes with distinct clonal dynamics. This diversity appears to be vital for shaping and maintaining bone tissue throughout life. We identified four major types of hSSCs, which exhibit over 90% transcriptional similarity. Interestingly, different variants of hSSCs are dispersed in distinct areas of the bone, providing the bone with a “positional memory” regarding its compositional architecture. The strategic interspersal of fibro-stromal and osteochondrogenic hSSCs in

specific ratios might be crucial for achieving the right balance between flexibility and rigidity in bone structure, similar to the use of rebars and cement in reinforced concrete. Moreover, we found a particular directionality in the clonal propagation of skeletal progenitors derived from hSSCs within a bone. This is likely critical for shaping the skeleton and determining its structural characteristics during development and growth. Disruptions in genes associated with major hSSC subtypes are often linked to genetic diseases.

One of our key observations was that fibroblast-skewed hSSCs tend to increase under various pathological conditions, an occurrence reminiscent of age-related myeloid skewing by HSCs.<sup>5</sup> This led us to consider whether the skewing of HSCs and hSSCs in humans is interconnected, perhaps through communication between the hematopoietic and skeletal compartments.<sup>19,20</sup> It would also be worthwhile to explore if certain point mutations, such as those found in clonal hematopoiesis of indeterminate potential (CHIP), could lead to the expansion of fibroblast-skewed hSSCs over time. Alternatively, these clones might already exist but become dominant during aging or in response to co-morbidities such as diabetes.<sup>17</sup> Fibroblast-skewed hSSCs are significantly less osteogenic than osteochondral hSSCs. Their pathological expansion results in poor ossification and bone healing. Understanding why aging promotes overexpansion of fibrogenic hSSCs could shed light on the interplay between aging, disease, and skeletal health. Given that fibroblast-skewed hSSCs expansion seems to be a common pathological mechanism in the skeleton, and fibrosis is a typical aging hallmark associated with altered fibroblast composition, our identification of diverse hSSC subtypes defined by distinct but tunable gene regulatory networks offers a new perspective for designing effective strategies against skeletal aging and disease.<sup>47</sup>

## Limitations of the study

Our studies of human tissues are bound by ethical considerations and limited by the available specimens. Here, we collected samples from fetal donors at gestational ages 17–20 weeks to capture the role of hSSCs during the early establishment of the skeleton. Similarly, our patient cohort specimens came from skeletally mature healthy and diseased donors between 18 and 77 years of age, mostly collected from fracture sites. Future work will have to fill the gap by adding hSSC data from a broader spectrum of fetal stages and intact adult skeletal tissue, e.g., warm autopsies, as well as pre-pubertal to adolescent age groups to achieve a comprehensive compendium of hSSCs throughout all life stages.

Given the need for prospective isolation of human cell populations for functional assessment and the lack of elegant *in vivo* genetic perturbation approaches commonly possible in murine models, we draw our conclusion from a collective of multiple affirming readouts. Refinement of *in vivo* clonal tracking strategies allowing simultaneous 3D multiplexed niche analysis will pave the way for a more granular understanding of hSSC lineage dynamics *in vivo*. Our generated single-cell datasets, in combination with related data generated by other groups, will also allow for the identification of additional markers that could further purify our already exceedingly homogeneous hSSC populations and their lineage tree.

## RESOURCE AVAILABILITY

### Lead contact

Further information and requests for resources and reagents should be directed to the lead contact, T.H. Ambrosi (thambrosi@ucdavis.edu).

### Materials availability

This study did not generate any unique reagents.

### Data and code availability

All scRNA-seq raw data and count files were deposited in the Gene Expression Omnibus as a SuperSeries under accession number GEO: GSE234040. Publicly available data used in this manuscript were sourced from SRA: PRJNA478935 and as specified in figures and figure legends. All code for new analyses pipelines presented in this manuscript has been deposited at GitHub for re-creating analyses (<https://github.com/sahoo00/Invariant-Circuits>, <https://doi.org/10.5281/zenodo.14823193>, v1.0.1) and <https://github.com/sahoo00/Clonetracker>, <https://doi.org/10.5281/zenodo.14823191>, v1.0.1).

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## AUTHOR CONTRIBUTIONS

T.H.A. and C.K.F.C. conceived the study. T.H.A. performed the majority of experiments and analyzed results. T.H.A., M.T.L., D.S., and C.K.F.C. wrote the manuscript. S.T., R.S., L.H.G., M.P.M., H.M.S., M.Y.H., Yuting Wang, E.J.H., F.F., K.C.W., L.S.K., D.D.L., J.S., L.Z., K.C., S.D.C., W.-J.L., M.M., N.F.N., N.L.V.R., E.E.W., J.K.L., A.S., Yongheng Wang, A.W., G.P.Y., S.G., J.A.B., M.J.G., and D.C.W. performed experiments and/or provided expertise. T.H.A., I.L.W., M.T.L., D.S., and C.K.F.C. supervised the project.

## DECLARATION OF INTERESTS

The authors (T.H.A., R.S., I.L.W., M.T.L., D.S., and C.K.F.C.) have filed an invention disclosure (USPTO Application #63/676,773; filed on July 29, 2024) related to this work entitled: “Compositions and methods for re-activation of dysfunctional skeletal stem cells.”

## STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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## SUPPLEMENTAL INFORMATION

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## STAR★METHODS

### KEY RESOURCES TABLE

| REAGENT or RESOURCE                                  | SOURCE                                            | IDENTIFIER                                                 |
|------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------|
| <b>Antibodies</b>                                    |                                                   |                                                            |
| Mouse anti-human CD45                                | BioLegend                                         | Cat#304029-BL; RRID: AB_2174123                            |
| Mouse anti-human CD235a                              | BioLegend                                         | Cat#306612-BL; RRID: AB_2116241                            |
| Mouse anti-human CD31                                | Thermo Fisher Scientific                          | Cat#13-0319-82; RRID: AB_466423                            |
| Mouse anti-human CD202b (TIE-2)                      | BioLegend                                         | Cat#334204; RRID: AB_1186046                               |
| Mouse anti-human CD146                               | BioLegend                                         | Cat#342010; RRID: AB_10644009                              |
| Mouse anti-human PDPN                                | Thermo Fisher Scientific                          | Cat#17-9381-42; RRID: AB_10801951                          |
| Mouse anti-human CD164                               | BioLegend                                         | Cat#324808; RRID: AB_2072587                               |
| Mouse anti-human CD73                                | BioLegend                                         | Cat#344016; RRID: AB_2561809                               |
| Streptavidin (APC-Alexa Fluor™ 750)                  | Thermo Fisher Scientific                          | Cat#SA1027; RRID: AB_2716627                               |
| Anti-alpha smooth muscle Actin                       | Abcam                                             | Cat# ab7817; RRID: AB_262054                               |
| Goat anti-mouse AF594                                | Abcam                                             | Cat# ab150120; RRID: AB_2631447                            |
| Mouse anti-human PDPN                                | eBioscience                                       | Cat#BMS1105; RRID: AB_10598795                             |
| Goat anti-mouse AF488                                | Abcam                                             | Cat# ab150113; RRID: AB_2687506                            |
| Mouse anti-Human Nuclear Antigen (HNA) antibody      | Abcam                                             | CAT#ab191181; RRID: AB_2885016                             |
| Rabbit anti-human Osteocalcin                        | Abcam                                             | CAT#ab93876; RRID: AB_10675660                             |
| Rabbit anti-human Collagen II                        | Abcam                                             | CAT#ab34712; RRID: AB_731688                               |
| RNAscope™ Probe- Hs-HIC1                             | Advanced Cell Diagnostics                         | Cat#470571                                                 |
| RNAscope™ Probe- Hs-SKIL-C3                          | Advanced Cell Diagnostics                         | Cat#427981-C3                                              |
| RNAscope™ Probe- Hs-SPP1-C2                          | Advanced Cell Diagnostics                         | Cat#420101-C2                                              |
| RNAscope™ Probe- Hs-FMO1-C2                          | Advanced Cell Diagnostics                         | Cat#892031-C2                                              |
| TaqMan® Gene Expression Probe GAPDH                  | Thermo Fisher Scientific                          | Hs02786624_g1                                              |
| TaqMan® Gene Expression Probe SPP1                   | Thermo Fisher Scientific                          | Hs00959010_m1                                              |
| TaqMan® Gene Expression Probe FBN1                   | Thermo Fisher Scientific                          | Hs00171191_m1                                              |
| TaqMan® Gene Expression Probe HAPLN1                 | Thermo Fisher Scientific                          | Hs01091999_m1                                              |
| TaqMan® Gene Expression Probe FMO1                   | Thermo Fisher Scientific                          | Hs01032912_m1                                              |
| TaqMan® Gene Expression Probe BMPR1A                 | Thermo Fisher Scientific                          | Hs04980288_g1                                              |
| TaqMan® Gene Expression Probe IHH                    | Thermo Fisher Scientific                          | Hs00745531_s1                                              |
| TaqMan® Gene Expression Probe SOX9                   | Thermo Fisher Scientific                          | Hs00165814_m1                                              |
| TaqMan® Gene Expression Probe TAGLN                  | Thermo Fisher Scientific                          | Hs01038777_g1                                              |
| <b>Biological samples</b>                            |                                                   |                                                            |
| Human fetal samples                                  | certified procurement services                    | Gestational age and sex listed in <a href="#">Table S1</a> |
| Adult samples                                        | Stanford University and UC Davis Health Hospitals | Age and sex listed in <a href="#">Table S1</a> .           |
| <b>Chemicals, peptides, and recombinant proteins</b> |                                                   |                                                            |
| Type II collagenase                                  | Sigma-Aldrich                                     | Cat#C6885                                                  |
| DNase I                                              | Worthington                                       | Cat#NC9199796                                              |
| PBS                                                  | GIBCO                                             | Cat#C14190500BT                                            |
| UltraPure™ 0.5M EDTA                                 | Invitrogen                                        | Cat#15575020                                               |
| DAPI and Hoechst Nucleic Acid Stains                 | Invitrogen                                        | Cat#D1306                                                  |
| MEM-alpha medium                                     | Fisher Scientific                                 | Cat#12561-056                                              |
| Human platelet-derived lysate                        | Stem Cell Technologies                            | Cat#06960                                                  |
| Penicillin-Streptomycin solution                     | Thermo Fisher Scientific                          | Cat#15140-122                                              |
| Crystal Violet                                       | Thermo Fisher Scientific                          | Cat#B21932.22                                              |
| Paraformaldehyde                                     | Thermo Fisher Scientific                          | Cat#047340.9L                                              |

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| REAGENT or RESOURCE                                                 | SOURCE                    | IDENTIFIER                                                                                                                                                                                                           |
|---------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dexamethasone                                                       | Thermo Fisher Scientific  | Cat#A17590.03                                                                                                                                                                                                        |
| Sodium b-glycerophosphate                                           | Sigma-Aldrich             | Cat#G9891                                                                                                                                                                                                            |
| L-Ascorbic acid 2-phosphate                                         | Sigma-Aldrich             | Cat#A8960                                                                                                                                                                                                            |
| Alizarin Red S                                                      | Sigma-Aldrich             | Cat#A5533                                                                                                                                                                                                            |
| DMH1                                                                | Selleck                   | Cat#S7146                                                                                                                                                                                                            |
| SAG 21k                                                             | Biotechnie                | Cat#5282                                                                                                                                                                                                             |
| Acetic Acid                                                         | Thermo Fisher Scientific  | Cat#A35-500                                                                                                                                                                                                          |
| Methanol                                                            | Fisher Scientific         | Cat#A412-4                                                                                                                                                                                                           |
| Alcian Blue                                                         | Sigma-Aldrich             | Cat#A5268                                                                                                                                                                                                            |
| Human Transforming growth factor β1                                 | Peprotech                 | Cat#100-21C                                                                                                                                                                                                          |
| Ethylenediaminetetraacetic acid (EDTA)                              | Invitrogen                | Cat#15573-038                                                                                                                                                                                                        |
| Bovine Serum Albumin                                                | Sigma-Aldrich             | Cat#A2153                                                                                                                                                                                                            |
| Triton X-100                                                        | Thermo Fisher Scientific  | Cat#A16046.AP                                                                                                                                                                                                        |
| Matrigel                                                            | Corning                   | Cat#354230                                                                                                                                                                                                           |
| Anorganic cancellous bone graft granulate                           | InterOss®                 | Cat#IOSG050                                                                                                                                                                                                          |
| Opti-MEM™ I Reduced Serum Medium                                    | Thermo Fisher Scientific  | Cat#31985070                                                                                                                                                                                                         |
| Sucrose                                                             | Fisher Scientific         | Cat#S5-500                                                                                                                                                                                                           |
| Trizol                                                              | Thermo Fisher Scientific  | Cat#15596026                                                                                                                                                                                                         |
| Nuclease-free water                                                 | Thermo Fisher Scientific  | Cat#10977023                                                                                                                                                                                                         |
| RNase inhibitor                                                     | Clontech                  | Cat#2313B                                                                                                                                                                                                            |
| dNTP                                                                | Invitrogen                | Cat#10297-018                                                                                                                                                                                                        |
| MVG sodium alginate                                                 | Pronova Novamatrix        | Cat#4200101                                                                                                                                                                                                          |
| Sodium periodate                                                    | Acros Organics            | CAS#7790-28-5                                                                                                                                                                                                        |
| CaCl2                                                               | MilliporeSigma            | CAS#10043-52-4                                                                                                                                                                                                       |
| Ammonium chloride                                                   | Sigma                     | Cat#A9434-500G                                                                                                                                                                                                       |
| Potassium bicarbonate                                               | Fisher Chemical           | Cat#P235-500                                                                                                                                                                                                         |
| <b>Critical commercial assays</b>                                   |                           |                                                                                                                                                                                                                      |
| RNAscope™ Multiplex Fluorescent Reagent Kit v2                      | Advanced Cell Diagnostics | Cat#323100                                                                                                                                                                                                           |
| Taq PCR Master Mix Kit                                              | Qiagen                    | Cat#201445                                                                                                                                                                                                           |
| QuantiTect Reverse Transcription Kit                                | Qiagen                    | Cat#205313                                                                                                                                                                                                           |
| CosMx™ Human Universal Cell Characterization Panel (RNA, 1000 Plex) | Nanostring                | CMX-H-USCP-1KP-R                                                                                                                                                                                                     |
| ERCC (External RNA Controls Consortium) ExFold RNA Spike-In Mix 2   | Invitrogen                | Cat#4456739                                                                                                                                                                                                          |
| SMARTScribe reverse transcriptase                                   | Clontech                  | Cat#639538                                                                                                                                                                                                           |
| KAPA HiFi HotStart ReadyMix                                         | Kapa Biosystems           | Cat#KK2602                                                                                                                                                                                                           |
| AMPure XP beads                                                     | Beckman Coulter           | Cat#A63882                                                                                                                                                                                                           |
| Nextera XT kit                                                      | Illumina                  | Cat#FC-131-1096                                                                                                                                                                                                      |
| Fragment Analyzer DNA/NGS Kits                                      | Agilent                   | DNF-474-0500                                                                                                                                                                                                         |
| Bioanalyzer High Sensitivity DNA kit                                | Agilent                   | 5067-4626                                                                                                                                                                                                            |
| 10X Chromium Next GEM Single Cell 3' v3.1 GEM kit                   | 10X Genomics Inc          | PN-1000121                                                                                                                                                                                                           |
| <b>Deposited data</b>                                               |                           |                                                                                                                                                                                                                      |
| Single-cell RNA sequencing data                                     | This paper                | NCBI Gene Expression Omnibus (GEO) repository, GEO: GSE234040                                                                                                                                                        |
| Clonetracker                                                        | Github                    | <a href="https://github.com/sahoo00/Clonetracker">https://github.com/sahoo00/Clonetracker</a> ( <a href="https://doi.org/10.5281/zenodo.14823191">https://doi.org/10.5281/zenodo.14823191</a> , v1.0.1)              |
| Invariant-Circuits                                                  | Github                    | <a href="https://github.com/sahoo00/Invariant-Circuits">https://github.com/sahoo00/Invariant-Circuits</a> ( <a href="https://doi.org/10.5281/zenodo.14823193">https://doi.org/10.5281/zenodo.14823193</a> , v1.0.1)) |

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| Continued                                                              |                     |                                                                                                                             |
|------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| REAGENT or RESOURCE                                                    | SOURCE              | IDENTIFIER                                                                                                                  |
| Experimental models: Organisms/strains                                 |                     |                                                                                                                             |
| NSG (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ) mice                            | JAX                 | JAX: 005557                                                                                                                 |
| Oligonucleotides                                                       |                     |                                                                                                                             |
| oligo dT30VN                                                           | IDT                 | custom: 5'-AAGCAGTGGTATCAACGCAGA GTACT30VN-3'                                                                               |
| Template-switching oligonucleotide (TSO)                               | Exiqon              | custom: 5'-AAGCAGTGGTATCAACGCAGA GTACATrGrG+G-3'                                                                            |
| IS PCR primers                                                         | IDT                 | custom: 5'-AAGCAGTGGTATCAACGCAG AGT-3'                                                                                      |
| Partial Read 1 primer (CloneTracker assay)                             | IDT                 | 5'-CTACACGACGCTCTTCCGATCT-3'                                                                                                |
| Adaptor-FBP1 (CloneTracker assay)                                      | IDT                 | 5'-GTGACTGGAGTTGAGACGTGTGCTCTT CCGATCTCCGCCACCGAACGCAACGCA CGCA-3'                                                          |
| Recombinant DNA                                                        |                     |                                                                                                                             |
| CloneTracker XP™ 10M Barcode-3' Library in pScribe5-Venus-Puro plasmid | Collecta            | Cat#BCXP10M3VP-P                                                                                                            |
| Software and algorithms                                                |                     |                                                                                                                             |
| Scanpy                                                                 | Github              | v1.9, <a href="https://github.com/scverse/scanpy">https://github.com/scverse/scanpy</a>                                     |
| Squidpy                                                                | Github              | v1.2.1, <a href="https://github.com/scverse/squidpy">https://github.com/scverse/squidpy</a>                                 |
| CellphoneDB                                                            | Github              | v5, <a href="https://github.com/Teichlab/cellphonedb">https://github.com/Teichlab/cellphonedb</a>                           |
| CellCharter                                                            | Github              | v0.3.0, <a href="https://github.com/CSOgroup/cellcharter">https://github.com/CSOgroup/cellcharter</a>                       |
| bcl2fastq2                                                             | Github              | v2.18, <a href="https://github.com/bbrwnj/bcl2fastq">https://github.com/bbrwnj/bcl2fastq</a>                                |
| Skewer                                                                 | Github              | v0.2.2, <a href="https://github.com/relipmoc/skewer">https://github.com/relipmoc/skewer</a>                                 |
| STAR aligner                                                           | Github              | v2.6.1d, <a href="https://github.com/alexdobin/STAR">https://github.com/alexdobin/STAR</a>                                  |
| RSEM                                                                   | Github              | v1.2.21, <a href="https://github.com/deweylab/RSEM">https://github.com/deweylab/RSEM</a>                                    |
| Cellranger                                                             | 10X Genomics Inc    | v7.1.0                                                                                                                      |
| dnacust                                                                | Github              | <a href="https://github.com/hsmurali/SCRIPT/tree/main/dnacust">https://github.com/hsmurali/SCRIPT/tree/main/dnacust</a>     |
| Scvelo                                                                 | Github              | v0.2.5, <a href="https://github.com/theislabs/scvelo">https://github.com/theislabs/scvelo</a>                               |
| StepMiner                                                              | Github              | <a href="https://github.com/suryatheja/StepMiner-Bioinformatics">https://github.com/suryatheja/StepMiner-Bioinformatics</a> |
| BooleanNet                                                             | Github              | <a href="https://github.com/sahoo00/BooleanNet">https://github.com/sahoo00/BooleanNet</a>                                   |
| MIDReG                                                                 | UCSD                | <a href="http://hegemon.ucsd.edu/microarray/MIDReG/">http://hegemon.ucsd.edu/microarray/MIDReG/</a>                         |
| Boolean Equivalent Correlated Clusters                                 | UCSD                | <a href="http://hegemon.ucsd.edu/Misc/biomarkers/bone.php?go=hs">http://hegemon.ucsd.edu/Misc/biomarkers/bone.php?go=hs</a> |
| GraphPad Prism                                                         | GraphPad Software   | v9.5.1, <a href="http://www.graphpad.com/scientific-software/prism">http://www.graphpad.com/scientific-software/prism</a>   |
| FlowJo                                                                 | FLOWJO LLC          | <a href="https://www.flowjo.com/">https://www.flowjo.com/</a>                                                               |
| Cytotrace                                                              | Stanford University | <a href="https://cytotrace.stanford.edu/">https://cytotrace.stanford.edu/</a>                                               |
| Enrichr                                                                | Github              | <a href="https://github.com/wjawaid/enrichR">https://github.com/wjawaid/enrichR</a>                                         |
| Biorender                                                              | Biorender           | <a href="https://www.biorender.com/">https://www.biorender.com/</a>                                                         |

## EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

### Fetal and adult human samples

The gestational age and sex of fetal specimens are listed in [Table S1](#). Six human fetal samples were obtained from certified procurement services and shipped overnight. Samples ranged in age from 17 to 20 weeks of gestation with no restrictions on race or gender. Adult hSSCs for functional and scRNAseq assays were obtained from fracture callus (day 1-8 after injury), nonunions (120-300 days after injury), uninjured periosteum, and Fibrous Dysplasia bone tissues. Age and sex of bone specimens are also listed in [Table S1](#). Fracture callus/nonunion/periosteal/Fibrous Dysplasia specimens were obtained from Stanford University and UC Davis Hospitals. Fetal sample procurement and handling as well as collection of all other tissues, including fracture callus, nonunion and Fibrous

Dysplasia, used in this study were conducted under IRB-35711 (Stanford) and IRB-1997852 (UC Davis) in accordance with the guidelines and regulations set by the Institutional Review Board of Stanford University School of Medicine and UC Davis School of Medicine, respectively. On these IRBs, informed consent was not required as samples were considered biological waste. Briefly, fracture callus tissue was collected from patients undergoing an open approach and direct reduction of fracture fragments. Hematoma and fracture callus tissue that disturbed satisfactory alignment of displaced fragments was removed and saved for research purposes at time of surgery. Only tissue between the two ends of the fracture was removed to reassemble fragments prior to fixation. Intramedullary sampling was not included in this study. No restrictions were made regarding the race, gender, or age of the specimen's donor. Following excision, all specimens were placed on ice, and hSSCs were isolated as described below.

### Mice

Three-month old male immunodeficient NSG mice (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ; JAX: 005557) were used for transplantation studies. Mice were maintained at the Stanford University and UC Davis research animal facilities in accordance with university guidelines. Animals were given food and water ad libitum and housed in temperature- and light-controlled micro-insulators. All studies were approved via IACUC by the respective institutions.

## METHOD DETAILS

### Human fetal and adult bone processing

Human specimens were processed for the generation of a single cell solution as previously described.<sup>14</sup> Briefly, human fetal bones were separated and gently denuded of soft tissue using scissors and paper towels. Intact long bones, spine, pelvis and cranial tissues were collected. Long bones were additionally dissected by first gently stripping periosteal/perichondral tissue from the intact skeletal element using a razor blade and fine forceps. The cartilaginous bone ends were then removed to analyze the growth plate region. For spine tissue vertebral bodies from cerebral, thoracic and lumbar vertebrae were combined. Disc tissues were then separated from the rest of the vertebral body by applying force from both sides of the tissue using forceps at the intersection of disc and bone areas. For cranial tissue dissection, sutures were first removed using a razor blade and fine scissors. Frontal and parietal skull elements were then separated along the areas where sutures were removed. For tendon tissue, achilles and patella tendon fibers were combined. Once all the desired skeletal site tissues were dissected and cleaned, they were placed on ice in individual 10 cm Petri dishes containing FACS buffer. From adult human bone specimens regions containing distinguishable areas of clotted blood were dissected out with scalpels and the remaining tissue was then minced with razor blades, resuspended in 3000 U/mL type II collagenase (Sigma-Aldrich, Cat#C6885) digestion buffer supplemented with 100 U/mL DNase I (Worthington, Cat#NC9199796) and incubated at 37°C for 60 min under constant agitation. The supernatant was filtered through a 70 mm nylon mesh and quenched with staining buffer (2% fetal bovine serum (FBS), in phosphate-buffered saline (PBS), Gibco, Cat#C14190500BT). Cells were centrifuged at 200 g at 4°C followed by resuspension in the staining buffer.

### Fluorescence-activated cell sorting (FACS)

After removing supernatant from the derived pellet, human stromal cells were separated from red blood cells (RBCs) by addition of ACK 2 mL of ACK (Ammonium-Chloride-Potassium) lysis buffer. After three minutes incubation on ice 13 mL of staining buffer was added to quench lysis reaction followed by centrifugation at 200 g at 4°C. The resulting pellet was resuspended in 100 µL of staining buffer and stained with fluorochrome-conjugated antibodies against CD45 (BioLegend, Cat#304029-BL), CD235a (BioLegend, Cat#306612-BL), CD31 (Thermo Fisher Scientific, Cat#13-0319-82), CD202b (TIE-2) (BioLegend, Cat#334204), CD146 (BioLegend, Cat#342010), PDPN (Thermo Fisher Scientific, Cat#17-9381-42), CD164 (BioLegend, Cat#324808), and CD73 (BioLegend, Cat#344016). Secondary antibody staining against Biotin was performed after an extra wash step using Streptavidin (Thermo Fisher Scientific, Cat#SA1027). A FACS Aria II (BD Biosciences) or Aurora CS (CYTEK) was used for flow cytometric analysis and sorting of hSSCs. Gating was established with IgG controls and the help of fluorescence-minus-one (FMO: staining with all fluorophores except one) controls and negative DAPI staining was used as a measure for cell viability.

### Cell culture

Freshly sorted hSSCs were collected in 200 µL of staining buffer. For assays investigating downstream cell populations, human Bone-Cartilage-Stromal Progenitors (hBCSPs) and osteoprogenitors (hOPs) were sorted into individual collection tubes as well. Cells were pelleted and resuspended in expansion media containing MEM- $\alpha$  medium (Fisher Scientific: Cat#12561-056) with 10% human platelet-derived lysate (Stem Cell Technologies, Cat#06960), 1% Penicillin-Streptomycin solution (Pen-Strep; Thermo Fisher Scientific, Cat#15140-122). Plated culture vessels were maintained in an incubator at 37°C with 5% CO<sub>2</sub>. Fracture callus sample specimens yielding less than 30 hSSCs were excluded from further functional assays.

### Fibroblast Colony-Forming-Unit (CFU-F) in vitro assay

For fetal CFU-F in vitro assays, 500 hSSCs were sorted and plated into 6-well plates and cultured for 10-14 days. At this time, specimens were fixed with 2% PFA and stained with 0.5% Crystal Violet solution. The number of colonies (>30 cells) per well was counted to assess percentage of CFU-F. Human fracture callus derived hSSCs were sorted and plated at maximum of 500 cells per culture

vessel into 10 cm culture dishes. Colonies were counted two weeks after seeding using a light microscopy. Cells were then lifted and used for in vitro differentiation assays. Samples yielding less than three colonies were excluded from in vitro differentiation assays.

#### Osteogenic in vitro differentiation

Freshly sorted cells expanded for 10–14 days were lifted by trypsin digestion and seeded into 96-well plates for osteogenic differentiation. After overnight incubation expansion media was substituted with osteogenic differentiation media (DMEM medium with 10% FBS, 1% Pen-Strep, 100 nM dexamethasone, 10 mM sodium b-glycerophosphate [Sigma-Aldrich, Cat#G9891], 2.5 mM ascorbic acid 2-phosphate [Sigma-Aldrich, Cat#A8960]). Media was changed every other day for 14 days. Cells then were fixed with 2% PFA and stained with 2% Alizarin Red S (Sigma-Aldrich, Cat#A5533) to assess osteogenic potential. For experiments testing the activation of specific pathways on osteogenic differentiation DMH1 (4  $\mu$ M; Selleck: Cat#4126) and/or SAG 21k (0.05  $\mu$ M; CAS#946002-48-8, provided by the Beachy lab at Stanford University) were added during each osteogenic media change. Plates were washed and dried before subsequently dissolving stain in 300  $\mu$ L of 20% Methanol / 10% Acetic Acid solution. After the complete liberation of staining, 80  $\mu$ L of each well was transferred into a new 96-well plate in duplicates and absorbance was measured at 450 nm using a Ultraspec 2100 UV/Visible Spectrophotometer (Biochrom, Harvard Bioscience).

#### Chondrogenic in vitro differentiation

For chondrogenic differentiation cells were cultured in a micromass. For that, a 5  $\mu$ L drop containing  $8 \times 10^5$  cells was placed into the center of a 24-well plate well. Cells were incubated at 37 °C for two hours with 5% CO<sub>2</sub> and then provided with 500  $\mu$ L of pre-warmed chondrogenic media (DMEMhigh with 10 % FBS, 100 nM Dexamethasone, 1  $\mu$ M L-ascorbic acid-2-phosphate and 10 ng/ml Human Transforming growth factor  $\beta$ 1 [Peprotech: Cat#100-21C]). Media was refreshed every other day for two weeks. The micromass was then fixed with 2% PFA and stained with 1% Alcian Blue (Sigma-Aldrich, Cat#A5268). Chondrogenic potential was assessed by spectrophotometrically measuring absorption at 595 nm.

#### Immunocytochemistry

Freshly sorted hSSCs were plated at a density of 1,000 cells per 96-well plate well and maintained in expansion media for two days, allowing to attach and expand. At day two culture medium was removed and cells were fixed with 2% PFA. Fixed cell cultures were then permeabilized with 0.1% Triton X-100 solution and blocked with 3% BSA in PBS. Antibody staining was conducted with primary human Anti-alpha smooth muscle Actin antibody (Abcam, Cat# ab7817) and secondary goat anti-mouse AF594 (Abcam, Cat# ab150120). For nuclear staining specimens were treated with DAPI.

#### RNA in situ hybridization

Fresh human fetal long bone specimens (17 weeks) were fixed overnight before decalcification using 400 mM Ethylenediaminetetraacetic acid (EDTA; Invitrogen, Cat#15573-038) in PBS (pH 7.2) at 4 °C for 2 weeks. Specimens were then cryopreserved and sectioned at 10 microns. Sections were then processed for RNA in situ hybridization using RNAscope® Multiplex Fluorescent Reagent Kit v2 according to manufacturer's protocol (Advanced Cell Diagnostics, Cat#323100). Sections were RNA-probed with *HIC1* (ACD, Cat# 470571), *SKIL* (ACD, Cat# 427981-C3), *SPP1* (ACD, Cat# 420101-C2) and *FMO1* (ACD, Cat# 892031-C2). Additional immunofluorescence was added by staining with mouse anti-human PDPN antibody (eBioscience, Cat#BMS1105) followed by secondary goat anti-mouse AF488 (Abcam, Cat# ab150113) as well as counterstaining with DAPI.

#### Subcutaneous xenograft transplant model

Freshly dissected fetal phalanges (gestational age 17–20 weeks) were transplanted into the the dorsum of 3-month-old immunodeficient NSG mice (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ; JAX: 005557) as described previously.<sup>14</sup> Briefly, a small skin incision was made in the dorsum of anesthetized NSG mice and fetal phalanges slid under the skin. Interrupted sutures were applied to close the incision, and transplants were allowed to engraft and grow for indicated time points.

#### Subcutaneous cell transplants

Freshly sorted patient-derived hSSCs were sorted and expanded to confluency.  $1 \times 10^6$  cells were mixed with 5  $\mu$ L of Matrigel (Corning; Cat#354230) and seeded on 20 mg anorganic cancellous bone graft granulat (InterOss®, 0.25–1mm, Cat#IOSG050) before incubation at 37 °C for 1h. For experiments testing the activation of specific pathways DMH1 (2  $\mu$ g/mL; Selleck: Cat#4126) and SAG 21k (30 ng/mL; CAS#946002-48-8, provided by the Beachy lab at Stanford University) or PBS as control were added during the Matrigel/InterOss/Cell mixture process. Cell mixtures were placed at room and then immediately subcutaneously transplanted into the dorsum of 3-month old immunodeficient NSG mice (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ; JAX: 005557). Grafts were harvested at indicated timepoints and analyzed by microCT and histology.

#### In vivo clonal barcoding of hSSCs

For clonal tracking of single hSSCs,  $1.5 \times 10^5$  freshly sorted cells were seeded into 12-well plate with 1 mL of Opti-MEM Reduced Serum Media containing lentiviral barcode particles generated from CloneTracker XP™ 10M Barcode-3' Library in pScribe5-Venus-Puro plasmid (Collecta, Cat#BCXP10MSVP-P). Cells were spun in the well plate at 1,500g for 60 min at 24–27 °C. Directly following that step, 3 mL of fresh culture media was added, and cells were incubated at 37 °C for 24 h. At this point, the media was changed,

and cells allowed to further incubate for three days. Cells were then lifted using collagenase and stained with FACS antibodies. Sorted cells with phenotypic hSSC surface marker profile and simultaneous expression of Venus fluorophore were used for renal capsule transplantation assays into NSG mice. To prevent dilution of clones in final readout, 500 Venus<sup>+</sup>, barcoded hSSCs were mixed with 5,000 Venus<sup>-</sup>, non-barcoded hSSC supporter cells, mixed with 5  $\mu$ L of Matrigel and transplanted below the renal capsule of mice. Three transplants per hSSC source were conducted. Each mouse was transplanted with hSSCs from the growth plate under the renal capsule of one kidney and with hSSCs from the periosteum contralaterally. After three weeks, kidneys were excised from sacrificed mice and assessed for Venus expression in the graft tissue via microscope. Grafts were then dissected out, pooled by respective hSSC sources and processed for flow cytometric isolation of Venus<sup>+</sup> cells as described above. Sorted cells from grafts were processed for 10X genomics single cell RNA-sequencing for interrogation into cell fates and clonal relationships (described below).

#### Micro-computed tomography

Grafts were dissected from mice and fixed in 2% PFA overnight. Samples were then scanned using a Bruker Skyscan 1276 (Bruker Preclinical Imaging) with a source voltage of 85 kV, a source current of 200  $\mu$ A, a filter setting of Al 1 mm, and pixel size of 12 microns at 2016 x 1344. Reconstructed samples were analyzed using CT Analyser and CTvox software (Bruker).

#### Bi-cortical fracture model with hSSC transplants

FACS-purified primary hSSCs were processed from adult human donors and cultured in hSSC media until confluent. Cells were then trypsinized and loaded onto alginate scaffolds. Specifically, MVG sodium alginate was partially oxidized to a theoretical extent of 1% with sodium periodate, which created acetal groups susceptible to hydrolysis and then modified with arginine-glycine-aspartic acid peptide (RGD) to promote cell adhesion. First, Alginate was resuspended in sterile PBS (pH 7.25) at 2% (w/v) and optionally mixed with SAG 21k (Biotechne, CAT#5282, resuspended in DMSO at 40 mM) and/or DMH1 (Selleckchem, CAT#S7146, resuspended in DMSO at 40 mM) and mixed overnight at 4 °C. Cells were resuspended at  $5 \times 10^5$ /mL in the precursor solution and pipetted onto a silicone mold. To crosslink, samples were submerged in 20  $\mu$ L of 100 mM CaCl<sub>2</sub> for 20 minutes before transferring to a well plate. Each gel had a total volume of 6  $\mu$ L with final cell number of  $3 \times 10^4$  cells and optionally with 0.05  $\mu$ M SAG 21k and/or 4  $\mu$ M DMH1. Secondly, adult male NSG mice were anaesthetized using aerosolized isoflurane and analgesia was administered before incision. The femur was exposed after muscle distraction and lateral dislocation of the patella. A 25-gauge regular bevel needle (BD BioSciences) was inserted between the femoral condyles to provide relative intramedullary fixation, and a transverse fracture was created in the mid-diaphysis using micro-scissors. The gel detailed above was then placed adjacent to the fracture. Right after, the patella was relocated, and 6-0 nylon suture (Ethicon) was used to re-approximate the muscles. For flow cytometric analysis, mice were sacrificed after 10 days, and femurs were processed for analysis. For  $\mu$ CT and histology, mice were sacrificed after 14 days. Femurs were fixed in 2% paraformaldehyde overnight, decalcified in 10% EDTA, and dehydrated in 30% sucrose. Pentachrome and IHC staining were performed with anti-Human Nuclear Antigen (HNA) antibody (Abcam, Cat#ab191181), Anti-Osteocalcin antibody (Abcam, CAT#ab93876) and Anti-Collagen II antibody (Abcam, CAT#ab34712).

#### Quantitative PCR

For analysis of freshly isolated, non-cultured cell populations of interest, fetal cells were directly sorted into 1.5 mL Eppendorf tubes containing 500  $\mu$ L of Trizol. RNA was isolated from  $5 \times 10^4$  cells and generated cDNA was used for TaqMan Real-Time PCR Assays (ThermoFisher) with the following probes: *GAPDH*: Hs02786624\_g1; *SPP1*: Hs00959010\_m1; *FBN1*: Hs00171191\_m1; *HAPLN1*: Hs01091999\_m1; *FMO1*: Hs01032912\_m1. For analysis of in vitro factor treated hSSCs, primary cells were cultured in hSSC expansion media until confluent. Cells were then trypsinized and plated in a 12-well plate at 100,000 cells/well. Osteogenic media was added the day after plating, with or without the addition of the small molecules SAG21k (Hh activator) and DMH1 (BMP receptor inhibitor). After 3 days, cells were fixed with Trizol™ Reagent and processed for mRNA extraction. QuantiTect Reverse Transcription Kit was used to create the cDNA library, followed by the QuantiTect Taq PCR Master Mix Kit for qPCR. The following probes were used: *GAPDH*: Hs02786624\_g1; *BMPR1A*: Hs04980288\_g1; *IHH*: Hs00745531\_s1; *SOX9*: Hs00165814\_m1; *TAGLN*: Hs01038777\_g1.

#### Spatial transcriptomics with Nanostring CosMx

CosMx technology (Nanostring, Seattle, WA, USA), a high-plex spatial molecular imager, was employed to profile a 20-week human embryonic phalange specimen with the 1,000 genes CosMx Universal Cell Characterization Panel following the manufacturer's protocol. Briefly, fixed frozen specimens were prepared and sliced to a thickness of 5  $\mu$ m. PanCK, CD45, and CD68 antibodies and DAPI were used as morphological markers. Prepared samples were loaded onto the CosMx SMI instrument and underwent data collection, image processing, feature extraction, and cell segmentation. Transcriptomic profiles of individual cells were generated by integrating target transcript locations with cell segmentation information. These profiles were subsequently used for downstream data analysis. Spatial transcriptome analysis was performed using squidpy (v1.2.1) and scanpy (v1.9.3). Cells with <20 probe counts were filtered out. After quality control, 58,377 cells were obtained within the CosMx SP5 dataset of 104 fields of view (FOVs). To integrate cellular molecular features with tissue spatial histology, we employed CellCharter<sup>33</sup> as the algorithmic framework to identify, characterize and compare cellular niches. Five distinct cellular niches were annotated. Differentially expressed genes were obtained using the "rank\_genes\_groups" function in scanpy. An average of 113 genes per cell were detected for the periosteal region analyzed, 75

genes per cell for the growth plate region and 66 genes per cell for the whole phalange. CellphoneDB was used to investigate interactions between putative hSSCs and surrounding cells using only genes that were expressed in at least 5% of the cells within the respective areas analyzed. Cells located more than 200  $\mu\text{m}$  away were excluded from analysis.

#### Plate-based Smart-seq2 single cell RNA-sequencing

Single cell suspension derived from target tissues was processed for flow cytometric isolation as described above and published.<sup>14</sup> Single cell RNA-sequencing was conducted as reported before.<sup>19</sup> Briefly, single cells were index-sorted into 96-well plates containing 4  $\mu\text{L}$  lysis buffer [1  $\mu\text{L}$  RNase inhibitor (Clontech, Cat#2313B), 0.1% Triton (Thermo Fisher Scientific, Cat#85111), 2.5 mM dNTP (Invitrogen, Cat#10297-018), 2.5  $\mu\text{M}$  oligo dT30VN (IDT, custom: 5'-AAGCAGTGGTATCAACGCAGAGTACT30VN-3'), and 1:600,000 ERCC (External RNA Controls Consortium) ExFold RNA Spike-In Mix 2 (ERCC; Invitrogen, Cat#4456739) in nuclease-free water (Thermo Fisher Scientific, Cat#10977023)]. Immediately after sorting, plates were centrifuged at 3000  $\times$  g for 30 seconds at 4  $^{\circ}\text{C}$ , snap frozen on dry ice and kept at -80  $^{\circ}\text{C}$  until cDNA synthesis. Reverse transcription (RT) and cDNA pre-amplification was performed using the Smart-seq2 protocol with minor modifications. Lysis plates were thawed on ice and incubated at 72  $^{\circ}\text{C}$  for 3 minutes followed by immediate snap chilling on ice to anneal the oligo dT30VN primer. cDNA synthesis from single cell RNA was performed using oligo-dT primed reverse transcription with SMARTScribe reverse transcriptase (Clontech, Cat#639538) and a locked-nucleic acid containing template-switching oligonucleotide (TSO; Exiqon, custom: 5'-AAGCAGTGGTATCAACGCAGAGTACATrGrG+G-3'). PCR amplification was conducted using KAPA HiFi HotStart ReadyMix (Kapa Biosystems, Cat#KK2602) with IS PCR primers (IDT, custom: 5'-AAGCAGTGGTATCAACGCAGAGT-3'). To remove residual reaction components and oligos smaller than 400 base pairs, pre-amplified cDNA was purified using 0.65–0.75X volume of calibrated AMPure XP beads (Beckman Coulter, Cat#A63882). The cDNA concentration and size distribution for each well was assessed by a high-sensitivity AATI 96-capillary fragment analyzer (HS NGS Fragment Kit [1–6000bp]; Agilent). cDNA from each well was normalized to the desired concentration range (~0.15 ng/ $\mu\text{L}$ ) by dilution with EB buffer. The content of four 96-well plates was then consolidated into 384-well plates without cherry-picking/removing wells not holding any single cell cDNA. Subsequently the 384-well plates were used for library preparation (Nextera XT kit; Illumina, Cat#FC-131-1096) using a semi-automated pipeline. The barcoded libraries of each well were pooled, cleaned-up, and size-selected using two rounds (0.35x and 0.75x) of Agencourt AMPure XP beads (Beckman Coulter), as recommended by the Nextera XT protocol (Illumina). A high-sensitivity AATI fragment analyzer run was used to assess fragment distribution and concentrations. Pooled libraries were sequenced on a NovaSeq 6000 S4 flow cell (Illumina) to obtain 1–2 million 2  $\times$  150 base-pair paired-end reads per cell.

#### Single cell RNA-seq data processing

Data was demultiplexed using bcl2fastq2 2.18 (Illumina). Raw reads were further processed using skewer v0.2.2. for 3' quality-trimming, 3' adaptor-trimming, and removal of degenerate reads.<sup>48</sup> Trimmed reads were then aligned to the hg38 genome (Gencode version GRCh38.p13) with STAR aligner version 2.6.1d using 2-pass mapping,<sup>49</sup> and gene counts were calculated using RSEM 1.2.21.<sup>50</sup> When aligning female data the entire Y chromosome was masked, while only the PAR regions on the Y chromosome were masked for male data. If sex was unknown PAR-masking was used. For data pre-processing gene count tables were combined with metadata using the Scanpy python package v1.9.1. For each dataset investigated, cells with less than 500 genes or 25,000 read counts as well as mitochondrial gene content higher than 15%, ERCC content higher than 30%, and ribosomal gene content higher than 25% were excluded. Scrublet was then used to detect and remove residual duplicates. The data were normalized using size factor normalization so that every cell had 10,000 read counts, were  $\log_2$  transformed, underwent cell cycle regression,<sup>51</sup> and were scaled to a maximum value of 10. To account for batch effects and proper data integration, Scanpy-integrated ComBat was applied for cells derived from separate sequencing runs followed by data scaling. Highly variable genes were computed using default parameters. We then performed principal component analysis with elbow plot visualization, computed the neighborhood graph, and clustered the data using the Leiden method. Cytotrace (<https://cytotrace.stanford.edu/>) was used to explore the developmental potential of single cells analyzed.<sup>52</sup> To plot and infer cellular state trajectories Force-directed graph drawing (ForceAtlas2) with PAGA (Partition-based graph abstraction) or RNA Velocity was applied to the single-cell RNA-seq data.<sup>53</sup> Enrichr was used to identify pathways based on top upregulated genes in the target populations.<sup>54</sup> A total of 5,764 high quality cells across all donors listed in Table S1 were included in the analysis with an average gene count of 3,262 for fetal and 2,225 for patient hSSCs across datasets. Step-by-step instructions to reproduce preprocessing and analysis of data are available for deposited data in the Gene Expression Omnibus database (see Resource Availability statement).

#### 10X Genomics single cell RNA-sequencing

Freshly collected fracture nonunion tissue was harvested and processed as described for tissue dissociation protocol above. The single cell solution was stained for flow cytometry and 3x10<sup>6</sup> PI CD45 CD235a<sup>+</sup> cells were sorted into a collection tube containing FACS buffer. Cells were then processed with 10X Chromium Next GEM Single Cell 3' GEM kit (10X Genomics Inc, v3.1) according to manufacturer's instruction to target 10,000 cells. Barcoded samples were demultiplexed, aligned to the human genome (GRCh38.p13), and UMI-collapsed with the Cellranger toolkit with standard settings and -force-cells=10,000 (version 7.1.0, 10X Genomics Inc) sequenced on an partial-lane Illumina NovaSeq platform. We used the Scanpy package (version 1.9.1.) to explore the data. First, quality filtering to exclude multiplets and cells of poor quality was conducted by only keeping cells with a gene count of >500 and < 4000 with less than 20% mitochondrial and 35% ribosomal gene content as well as running Scrublet leaving 9,301

high quality cells with an average gene count of 2,225 per cell. Genes expressed in less than 3 cells across all cells were also removed from downstream analysis. Data was log-normalized, cell cycle regressed and scaled for analysis. Dimensionality reduction and Leiden clustering as well as subclustering were conducted choosing parameters based on PCA elbow plot. All further analyses were conducted as described for Smart-seq2 above.

#### Clonal tracking with barcoded single cell readout

For clonal in vivo tracking of hSSCs, lentivirally barcoded hSSC-derived graft cell suspensions were processed according to standard 10X Chromium Next GEM Single Cell 3' GEM kit (10X Genomics Inc, v3.1). To amplify clonal barcode sequences (Collecta, Cat#BCXP10M3VP-P) a separate PCR with cDNA after the amplification step (before ligation) was performed, since the 10X protocol only requires ~25% of cDNA for the standard protocol. The extra 10  $\mu\text{L}$  of cDNA were amplified using a combination of Partial Read 1 primer (5'-CTACACGACGCTCTTCGATCT-3') and an adaptor-FBP1 (Read 2 primer + barcode recognition site) primer (5'-GTGACTGTGAAGTTTCAGACGTGTGCTCTTCGATCTCCGACCAACGCAACGCACGCA-3'). Barcode amplification was run as recommended for cDNA reaction (10X protocol step 2.2) plus 4 extra cycles. From there the product was handled as per standard protocol starting from the index primer reaction without any fragmentation or ligation steps. Amplicon libraries were then sequenced via MiSeq. After demultiplexing, 10X barcodes and clonal barcodes were read out and matched to 10X barcoded cells in the regular analysis as described and deposited at GitHub (<https://github.com/sahoo00/Clonetracker>, <https://doi.org/10.5281/zenodo.14823191>, v1.0.1). Briefly, the FBP1 flanking sequence was used to locate the Clone Barcode ID in the reverse strand. Unique Cell barcode IDs were extracted from the first 28 nt-sequences of the forward strand. The extracted Clone Barcode ID reads were matched against the known Collecta barcode library and only reads with barcodes that matched the Collecta library were kept. The Cell Barcode ID and Clone Barcode ID were combined and stored in a fastq file. The fastq files were filtered using quality information. The first 16 nt-sequences from this fastq file represent the Cell barcode ID, while the last 48-nt sequences are the Clone Barcode ID. The Cell Barcodes and Clone Barcodes were clustered using the DNACLUSt tool. Scanpy (version 1.9.1.) was used and cells were quality filtered only keeping cells with a gene count of >500 and < 6000 with less than 20% mitochondrial and 35% ribosomal gene content as well as running Scrublet leaving 745 high quality cells (433 unique barcodes) of grafts derived from growth plate hSSCs and 509 high quality cells (343 unique barcodes) from periosteal hSSCs with an average gene count of 3,178 and 3,171 per cell, respectively. Clonal barcodes for all filtered cells were identified. Analysis was conducted as described above for 10X single cell RNA-sequencing data. We were able to recover 150 GP hSSCs and 85 PE hSSCs of the initial 3x500 transplanted hSSCs from each subregion. Cells were only counted as clones if the identical barcode appeared in at least two single cells of which at least one cell could be identified as a hSSC, yielding a recovery of 46 clones for growth plate hSSCs and 21 clones for periosteal hSSCs. For trajectory inference scvelo (version 0.2.5.) was used.

#### StepMiner for Boolean Analysis

StepMiner is a computational tool that identifies step-wise transitions in a time-series data.<sup>55</sup> StepMiner performs an adaptive regression scheme to identify the best possible step up or down based on sum-of-square errors. The steps are placed between time points at the sharpest change between low expression and high expression levels, which gives insight into the timing of the gene expression-switching event. To fit a step function, the algorithm evaluates all possible step positions, and for each position, it computes the average of the values on both sides of the step for the constant segments. An adaptive regression scheme is used that chooses the step positions that minimize the square error with the fitted data. Finally, a regression test statistic is computed as follows:

$$F \text{ stat} = \frac{\sum_{i=1}^n (\hat{X}_i - \bar{X})^2 / (m - 1)}{\sum_{i=1}^n (X_i - \hat{X}_i)^2 / (n - m)}$$

Where  $X_i$  for  $i = 1$  to  $n$  are the values,  $\hat{X}_i$  for  $i = 1$  to  $n$  are fitted values.  $m$  is the degrees of freedom used for the adaptive regression analysis.  $\bar{X}$  is the average of all the values:  $\bar{X} = \frac{1}{n} * \sum_{i=1}^n X_i$ . For a step position at  $k$ , the fitted values  $\hat{X}_i$  are computed by using  $\frac{1}{k} * \sum_{i=1}^k X_i$  for  $i = 1$  to  $k$  and  $\frac{1}{(n-k)} * \sum_{i=k+1}^n X_i$  for  $i = k+1$  to  $n$ .

#### Discovery of Boolean Implication Relationships (BIRs)

Boolean logic is a simple mathematical relationship of two values, i.e., high/low, 1/0, or positive/negative. The Boolean analysis of gene expression data requires the conversion of expression levels into two possible values. The StepMiner algorithm is reused to perform Boolean analysis of gene expression data.<sup>56</sup> The Boolean analysis is a statistical approach which creates binary logical inferences that explain the relationships between phenomena. Boolean analysis is performed to determine the relationship between the expression levels of pairs of genes (Figures S6A and S6B). The StepMiner algorithm is applied to gene expression levels to convert them into Boolean values (high and low). In this algorithm, first the expression values are sorted from low to high and a rising step function is fitted to the series to identify the threshold. Middle of the step is used as the StepMiner threshold. This threshold is used to convert gene expression values into Boolean values. A noise margin of 2-fold change is applied around the threshold to determine intermediate values, and these values are ignored during Boolean analysis. In a scatter plot, there are four possible quadrants based on Boolean values: (low, low), (low, high), (high, low), (high, high). A Boolean implication relationship is observed if any one of

the four possible quadrants or two diagonally opposite quadrants are sparsely populated. Based on this rule, there are six kinds of Boolean implication relationships. Two of them are symmetric (corresponding to the positively correlated genes), opposite (corresponding to the highly negatively correlated genes). Four of the Boolean relationships are asymmetric, and each corresponds to one sparse quadrant: (low => low), (high => low), (low => high), (high => high). BooleanNet statistics is used to assess the sparsity of a quadrant and the significance of the Boolean implication relationships.<sup>56,57</sup> Given a pair of genes A and B, four quadrants are identified by using the StepMiner thresholds on A and B by ignoring the Intermediate values defined by the noise margin of 2-fold change (+/- 0.5 around StepMiner threshold). Number of samples in each quadrant are defined as  $a_{00}$ ,  $a_{01}$ ,  $a_{10}$ , and  $a_{11}$  which is different from X in the previous equation of F stat. Total number of samples where gene expression values for A and B are low is computed using the following equations.

$$nA_{low} = (a_{00} + a_{01}), nB_{low} = (a_{00} + a_{10}),$$

Total number of samples considered is computed using following equation.

$$total = a_{00} + a_{01} + a_{10} + a_{11}$$

Expected number of samples in each quadrant is computed by assuming independence between A and B. For example, expected number of samples in the bottom left quadrant  $e_{00} = \hat{n}$  is computed as probability of A low ( $(a_{00} + a_{01})/total$ ) multiplied by probability of B low ( $(a_{00} + a_{10})/total$ ) multiplied by total number of samples. The following equation is used to compute the expected number of samples where  $ij$  can be 00, 01, 10, 11.

$$n = a_{ij}, \hat{n} = (nA_{low} / total * nB_{low} / total) * total$$

To check whether a quadrant is sparse, a statistical test for ( $e_{00} > a_{00}$ ) or ( $\hat{n} > n$ ) is performed by computing  $S_{00}$  and  $p_{00}$  using following equations. A quadrant is considered sparse if  $S_{00}$  is high ( $\hat{n} > n$ ) and  $p_{00}$  is small.  $S_{ij}$  is used in the following equation where  $ij$  can be 00, 01, 10, 11.

$$S_{ij} = \frac{\hat{n} - n}{\sqrt{\hat{n}}}$$

$$p_{00} = \frac{1}{2} \left( \frac{a_{00}}{(a_{00} + a_{01})} + \frac{a_{00}}{(a_{00} + a_{10})} \right)$$

A suitable threshold is chosen for  $S_{ij} > sThr$  and  $p_{ij} < pThr$  to check sparse quadrant with an acceptable false discovery rate (FDR < 0.001). A Boolean implication relationship is identified when a sparse quadrant is discovered using the following equation.

**Boolean Implication** = ( $S_{ij} > sThr$ ,  $p_{ij} < pThr$ ) where  $ij$  can be 00, 01, 10, 11 (see below).

A relationship is called Boolean equivalent if top-left and bottom-right quadrants are sparse.

$$Equivalent = (S_{01} > sThr, p_{01} < pThr, S_{10} > sThr, p_{10} < pThr)$$

Boolean opposite relationships have sparse top-right ( $a_{11}$ ) and bottom-left ( $a_{00}$ ) quadrants.

$$Opposite = (S_{00} > sThr, p_{00} < pThr, S_{11} > sThr, p_{11} < pThr)$$

Boolean equivalent and opposite are symmetric relationship because the relationship from A to B is same as from B to A. Asymmetric relationship forms when there is only one quadrant sparse (A low => B low: top-left; A low => B high: bottom-left; A high => B high: bottom-right; A high => B low: top-right). These relationships are asymmetric because the relationship from A to B is different from B to A. For example, A low => B low and B low => A low are two different relationships.

A low => B high is discovered if the bottom-left ( $a_{00}$ ) quadrant is sparse and this relationship satisfies following conditions.

$$A \text{ low} \Rightarrow B \text{ high} = (S_{00} > sThr, p_{00} < pThr)$$

Similarly, A low => B low is identified if the top-left ( $a_{01}$ ) quadrant is sparse.

$$A \text{ low} \Rightarrow B \text{ low} = (S_{01} > sThr, p_{01} < pThr)$$

A high => B high Boolean implication is established if the bottom-right ( $a_{10}$ ) quadrant is sparse as described below.

$$A \text{ high} \Rightarrow B \text{ high} = (S_{10} > sThr, p_{10} < pThr)$$

Boolean implication A high => B low is found if the top-right ( $a_{11}$ ) quadrant is sparse using following equation.

$$A \text{ high} \Rightarrow B \text{ low} = (S_{11} > sThr, p_{11} < pThr)$$

For each quadrant a statistic  $S_{ij}$  and an error rate  $p_{ij}$  is computed.  $S_{ij} > sThr$  and  $p_{ij} < pThr$  are the thresholds used on the BooleanNet statistics to identify Boolean implication relationships where  $ij$  can be 00, 01, 10, 11.

Boolean analyses in the test dataset GSE119087 uses a threshold of  $sThr = 3$  and  $pThr = 0.1$ . False discovery rate is computed for these thresholds (FDR < 0.000001) by using randomly permuting gene expression data in GSE119087.

## Boolean Network Explorer (BoNE)

Boolean network explorer (*BoNE*) provides an integrated platform for the construction, visualization and querying of a network of progressive changes underlying a disease or a biological process in two steps (Figure S6C): First, the expression levels of all genes in these datasets were converted to binary values (high or low) using the StepMiner algorithm. Second, gene expression relationships between pairs of genes were classified into one-of-six possible Boolean Implication Relationships (BIRs), two symmetric and four asymmetric, and expressed as Boolean implication statements. This offers a distinct advantage from conventional computational methods (Bayesian, Differential, etc.) that rely exclusively on symmetric linear relationships in networks. The other advantage of using BIRs is that they are robust to the noise of sample heterogeneity (i.e., healthy, diseased, genotypic, phenotypic, ethnic, interventions, disease severity) and every sample follows the same mathematical equation, and hence is likely to be reproducible in independent validation datasets. In the resultant Boolean implication network, *BoNE* enables data science in both unsupervised and supervised way while remaining agnostic to the sample type.

## Boolean implication network construction

Boolean implication analysis was performed in the large diverse human dataset GSE119087 ( $n = 25,955$  human samples, Figures S6D and S6E). First, gene expression levels were converted to Boolean values (high and low) using StepMiner algorithm. The expression values are sorted from low to high and a rising step function is fitted to the series to identify the threshold. The relationship between two genes was evaluated in the context of high and low values. These relationships are called Boolean implication relationships (BIRs) because they are represented by logical implication ( $\Rightarrow$ ) formula. BooleanNet statistics is used to assess the significance of the Boolean implication relationships.  $S > 3$  and  $p < 0.1$  are the thresholds (False Discovery Rate < 0.0001) used on the BooleanNet statistics during the data analysis to identify Boolean implication relationships. We relaxed this threshold  $S > 3$  and  $p < 0.15$  to identify Boolean Equivalent relationships. A noise margin of 2-fold change is applied around the threshold to determine intermediate values and these values are ignored during Boolean analysis. StepMiner threshold is denoted in red color and the noise margin is denoted in light blue colors in scatterplots analyzing Boolean relationships between genes. The Boolean implication network contains the six possible Boolean relationships between genes in the form of a directed graph with nodes as genes and edges as the Boolean relationship between the genes. The nodes in the Boolean implication network (BIN) are genes and the edges correspond to BIRs. Equivalent and Opposite relationships are denoted by undirected edges and the other four types (low => low; high => low; low => high; high => high) of BIRs are denoted by having a directed edge between them.

## Boolean paths

The asymmetric BIRs provide a unique dimension to the network that is fundamentally different from any other gene expression networks in the literature. Traversing a set of nodes in a directed graph of the Boolean network constitutes a Boolean path that can be interpreted as follows. A simple Boolean path involves two nodes and the directed edge between them (Figure S6F). This simple Boolean path can be interpreted as follows. For the nodes X and Y with  $X \text{ low} \Rightarrow Y \text{ low}$  only quadrant  $\#(low, high)$  is sparse; the other quadrants are filled with samples. Assuming monotonicity in X and Y, the quadrants can be ordered in two possible ways: (low, low)-(high, low)-(high, high) and (high, high)-(high, low)-(low, low). The path corresponds to (low, low)-(high, low)-(high, high) begins with X low and Y low. This is interpreted as X turns on first and then Y turns on along a hypothetical biological path defined by the sample order. Similarly, Y turns off first and then X turns off in the path (high, high)-(high, low)-(low, low). A complex path in the Boolean network involves more than one Boolean implication relationship (Figures S6G and S6H). Three Boolean implication relationships can be used to group samples into five bins and the bins can be ordered in two possible ways (forward, reverse).

## Boolean Implication Network and BTR

A curated list of known bone processes-related genes was combined with selected genes of the top differentially expressed genes ( $p\text{-value} < 0.05$ ,  $\text{LogFC} > 1.16$  and  $< -1.16$ ; total of 139 genes) within each analyzed group (functional vs. dysfunctional hSSCs). Genes were grouped according to their expression for displaying results of the complex Boolean implication network (BIN). A complex BIN (Complex-BIN) was created by identifying all significant pairwise Boolean implication relationships (BIRs) derived from a large database (25,955 datasets) of publicly available microarrays of human tissues, diseases, and cell lines (GSE119087). A previously published BooleanNet algorithm was performed to determine the BIRs between genes. Briefly, the BooleanNet algorithm identifies the type of BIR by searching the sparsely populated quadrant(s) using a statistical threshold S and likelihood error rate p in a scatterplot between two genes. These relationships are called BIRs because they are represented by logical implication ( $\Rightarrow$ ) formula. There are six types of BIR: high => high, low => low, high => low, low => high, equivalent, and opposite. The first four are asymmetric and have only one sparse quadrant while the latter two are symmetric and have two sparse quadrants.  $S > 3$  and  $p < 0.15$  are the thresholds used on the BooleanNet statistics for symmetric BIRs. For asymmetric BIRs,  $S > 3$  and  $p < 0.10$  is used. The Complex-BINs are displayed in the form of a directed graph where edges (arrows) correspond to the observed BIRs between genes and nodes representing genes. To simplify and reduce the size of Complex-BIN, we used Boolean equivalent relationships to cluster genes. There are three zones in each Boolean Implication network (two zones for analyzed groups and one for the shared genes between those two groups). First, we found Boolean equivalent relationships within each zone, then we found Boolean equivalent relationships between different zones. Finally, we clustered all connected components between zones into one clustered-node in the shared zone (each member of the clustered-node is labeled as a shared gene regardless of their original group). We identified relationships between clustered-nodes and the remaining genes by combining the BIRs of members for each clustered-node to build the Clustered-BIN. Next, we selected

four BIRs (high=>high, high=>low, opposite and low=>low) to simplify the Clustered-BIN by traversing the entire graph by running a depth-first search to find all transitive complex pathways. Each complex pathway was formed from a chain of high=>high BIR, followed by a high=>low or opposite BIR followed by a chain of low=>low BIR. As the starting point a gene from one of the analyzed groups (zones) was chosen following a path through shared genes and ending in the opposite analyzed group (zone). The simple paths containing three BIRS of high=>high, high=>low and low=>low is interpreted in the context of the previously published MiDReG algorithm (Figures S6F–S6H). Ordering of the genes was performed using the asymmetric BIRs and MiDReG algorithm. The simplified Clustered-BIN helped to gain predicted orders of genes between analyzed groups. In the final step, all cells relevant to the analyzed groups and all genes in each selected complex pathway from the simplified Clustered-BIN were used in BoolTraineR (BTR), an algorithm for rebuilding and training the asynchronous Boolean model (Figure S6I). BTR uses a Boolean State Space (BSS) scoring function such that the predicted model can resemble a single-cell expression dataset. Furthermore, because of the conversion of expression levels to two possible values of 0/1, Boolean models are relatively robust to the presence of dropouts. The initial expression state (Boolean value) of each gene in the beginning of the selected pathways of MiDReG algorithm was used as an initial state for BTR model learning. We also used the BIRs between genes in each selected pathway as an initial Boolean model to start the search. Finally, BTR returned the closest predicted model based on the single-cell dataset. The code has been deposited at GitHub (<https://github.com/sahoo00/Invariant-Circuits>, [<https://doi.org/10.5281/zenodo.14823193>], v1.0.1).

#### Boolean Equivalent Correlated Clusters (BECC)

BECC was used to identify gene clusters that share invariant relationships with a seed gene.<sup>38</sup> We used *ACTA2* as a seed gene and a publicly available 25,955 diverse human samples downloaded from GEO (GSE119087) to identify and validate an invariant signature for fibroblast-like cells. The algorithm first computes a list of genes that have a Boolean Equivalent relationship with the seed gene. The list of genes was then expanded by performing Boolean analysis by following the Boolean equivalent relationship two more times. In the second step, a score for each gene in the final list was computed from the correlation coefficient and the slope of the fitted line comparing with the pre-defined seed gene. Finally, a threshold was imposed using the StepMiner algorithm to define a list of high-confidence genes with potentially superior performance than the seed gene. Once defined, all gene clusters were subsequently validated across global, bone tissues, and pathological bone diseases in human and murine datasets. This effort resulted in a ranked list of 33 probesets, corresponding to 25 unique genes, based on similarity to *ACTA2* (with probeID = 200974\_at). *TAGLN* and *MYL9* showed strong correlation patterns with the seed gene and each other in many validation datasets. The results of the BECC algorithm on human, bone-related and other species datasets can be explored via <http://hegemon.ucsd.edu/Misc/biomarkers/bone.php?go=hs>. We also used BECC to find invariant markers for osteogenic and chondrogenic hSSCs but no high-confidence gene could be found.

#### Quantification and statistical analysis

##### Statistics and reproducibility

All data presented were validated with multiple experimental replicates and plotted as stated in the figure or figure legend. Two-tailed Student's t-tests were performed with corrections for non-normality (Mann-Whitney test) or unequal variances (Welch's test) where appropriate, using GraphPad Prism v.9.5.1 (GraphPad Software). For more than two groups, one-way ANOVA with appropriate posthoc tests as indicated in the figure legends was used. No statistical method was used to predetermine sample size. No method of randomization was used. The investigators were partially blinded to allocation during experiments or outcome assessment.

## Supplemental Information

### Human skeletal development and regeneration

### are shaped by functional diversity of stem cells

### across skeletal sites

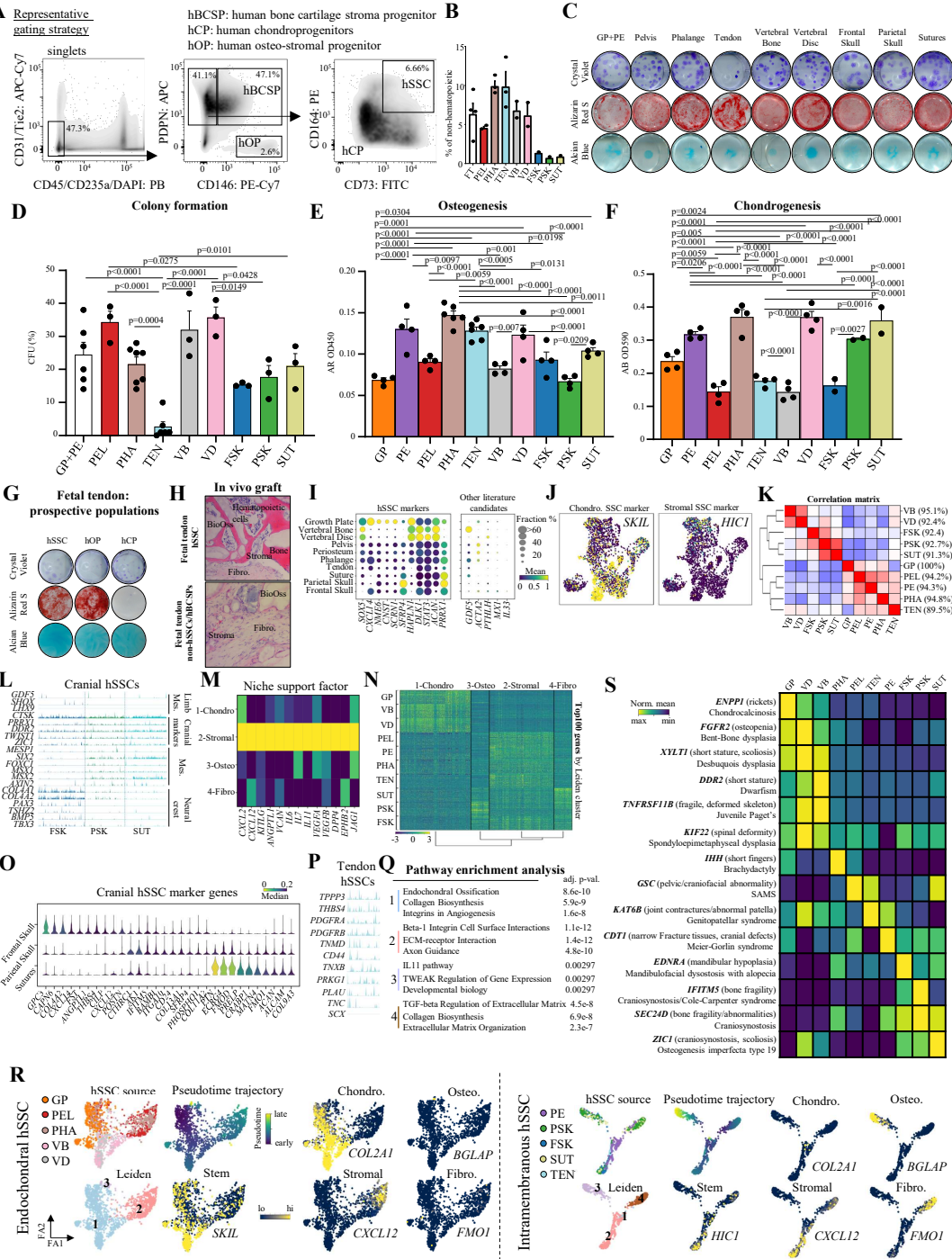
Thomas H. Ambrosi, Sahar Taheri, Kun Chen, Rahul Sinha, Yuting Wang, Ethan J. Hunt, L. Henry Goodnough, Matthew P. Murphy, Holly M. Steininger, Malachia Y. Hoover, Franco Felix, Kelly C. Weldon, Lauren S. Koepke, Jan Sokol, Daniel Dan Liu, Liming Zhao, Stephanie D. Conley, Wan-Jin Lu, Maurizio Morri, Norma F. Neff, Noelle L. Van Rysselberghe, Erika E. Wheeler, Yongheng Wang, J. Kent Leach, Augustine Saiz, Aijun Wang, George P. Yang, Stuart Goodman, Julius A. Bishop, Michael J. Gardner, Derrick C. Wan, Irving L. Weissman, Michael T. Longaker, Debashis Sahoo, and Charles K.F. Chan

**Table S1. Individual human skeletal specimens used in human skeletal stem cell scRNAseq analyses, Related to Figures 1-5.**

| #  | Sample                                      | Age              | Gender | scApproach   | Cell # |
|----|---------------------------------------------|------------------|--------|--------------|--------|
| 1  | Fetal 1                                     | 17 weeks (gest.) | Female | Smart-seq2   | 366    |
| 2  | Fetal 2                                     | 20 weeks (gest.) | Female | Smart-seq2   | 1340   |
| 3  | Fetal 3                                     | 18 weeks (gest.) | Female | Smart-seq2   | 870    |
| 4  | Fetal 4                                     | 17 weeks (gest.) | Female | Smart-seq2   | 885    |
| 5  | Fetal 5                                     | 17 weeks (gest.) | Female | 10X Genomics | 1,254  |
| 6  | Fetal 6                                     | 20 weeks (gest.) | Female | CosMx        | 58,377 |
| 7  | Fracture tissue 1                           | 18 years         | Female | Smart-seq2   | 88     |
| 8  | Fracture tissue 2                           | 25 years         | Male   | Smart-seq2   | 164    |
| 9  | Fracture tissue 3                           | 27 years         | Male   | Smart-seq2   | 23     |
| 10 | Fracture tissue 4                           | 28 years         | Male   | Smart-seq2   | 131    |
| 11 | Fracture tissue 5                           | 30 years         | Female | Smart-seq2   | 25     |
| 12 | Fracture tissue 6                           | 32 years         | Male   | Smart-seq2   | 46     |
| 13 | Fracture tissue 7                           | 34 years         | Female | Smart-seq2   | 71     |
| 14 | Fracture tissue 8                           | 39 years         | Female | Smart-seq2   | 91     |
| 15 | Fracture tissue 9                           | 53 years         | Male   | Smart-seq2   | 43     |
| 16 | Fracture tissue 10                          | 61 years         | Female | Smart-seq2   | 74     |
| 17 | Fracture tissue 11                          | 63 years         | Female | Smart-seq2   | 74     |
| 18 | Fracture tissue 12                          | 65 years         | Female | Smart-seq2   | 54     |
| 19 | Fracture tissue 13                          | 67 years         | Female | Smart-seq2   | 39     |
| 20 | Fracture tissue 14                          | 72 years         | Male   | Smart-seq2   | 241    |
| 21 | Fracture tissue 15                          | 77 years         | Female | Smart-seq2   | 79     |
| 22 | Femoral periosteum                          | 65 years         | Male   | Smart-seq2   | 32     |
| 23 | Nonunion 1                                  | 51 years         | Male   | Smart-seq2   | 43     |
| 24 | Nonunion 2                                  | 59 years         | Female | Smart-seq2   | 37     |
| 25 | Nonunion 2                                  | 60 years         | Female | Smart-seq2   | 107    |
| 26 | Nonunion 4                                  | 62 years         | Female | Smart-seq2   | 90     |
| 27 | Nonunion 5                                  | 63 years         | Male   | 10X Genomics | 9,301  |
| 28 | Fibrous Dysplasia (mandible/maxillary bone) | 30 years         | Female | Smart-seq2   | 49     |

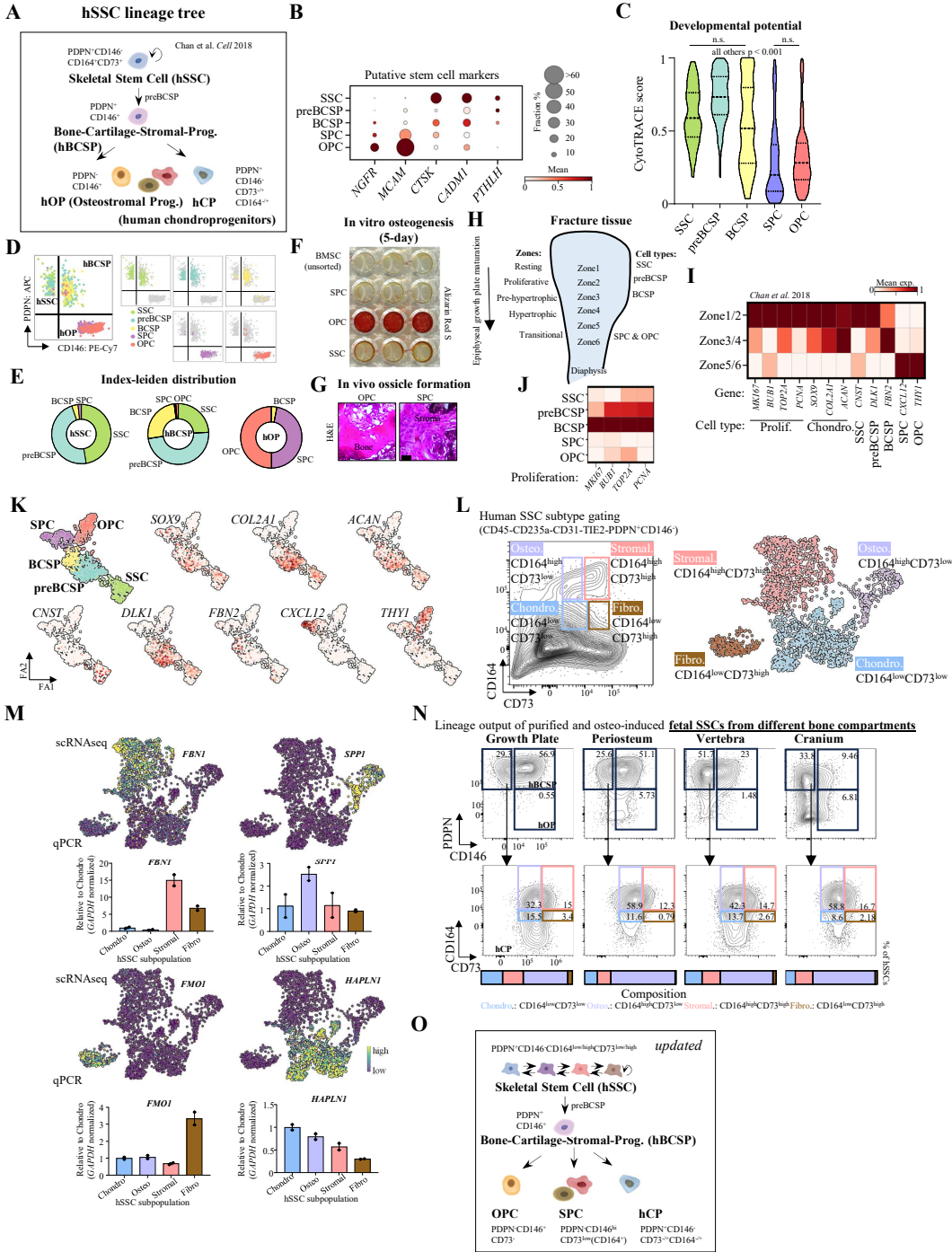
**Table S1. Individual human skeletal specimens used in human skeletal stem cell scRNAseq analyses, related to Figures 1-5.** Individual human skeletal specimens used in human single cell sequencing studies. (gest.: gestational age; sc: single cell)

**Figure S1. Skeletal site-specific diversity of hSSC, Related to Figure 1.**



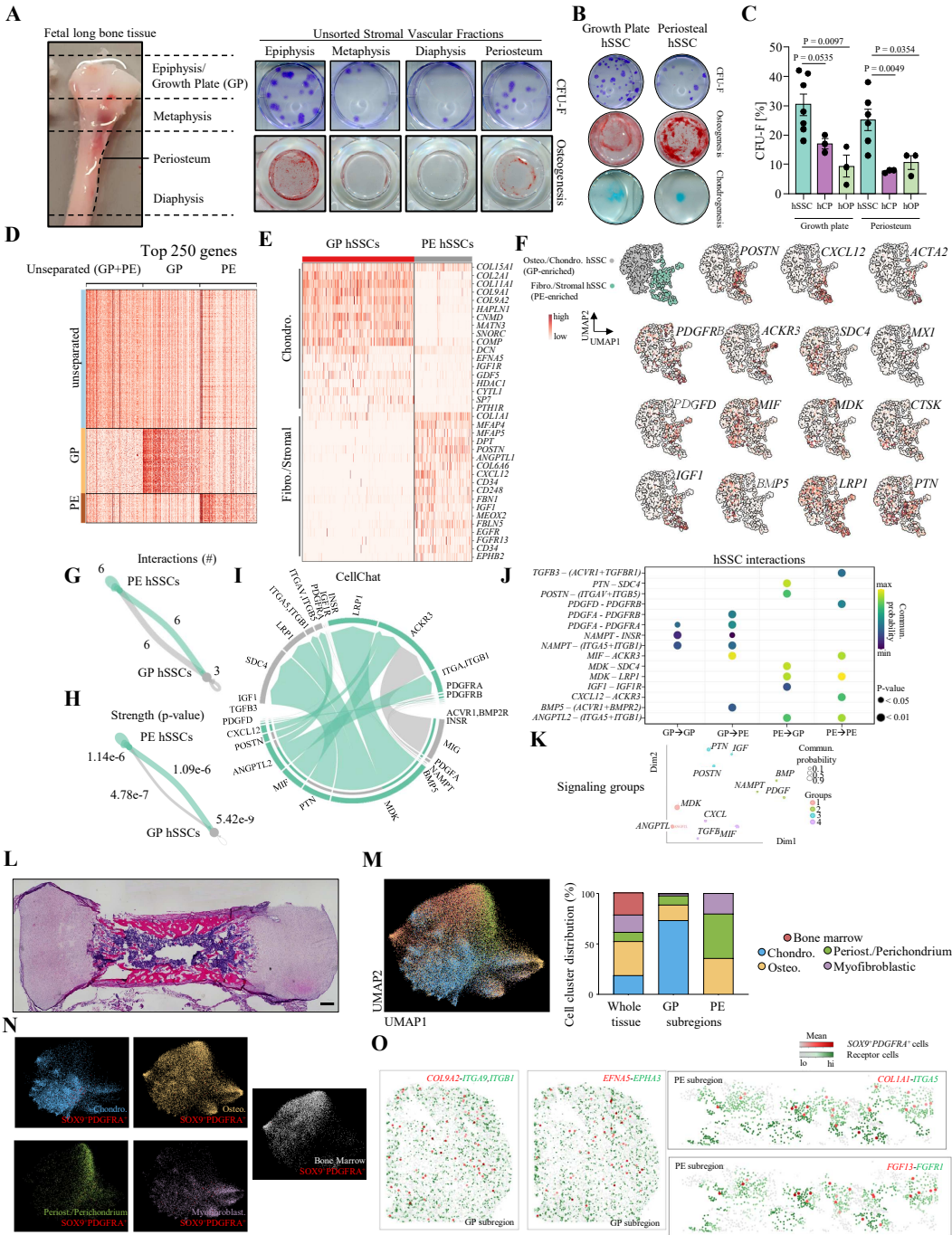
**Figure S1. Skeletal site-specific diversity of hSSC, Related to Figure 1.** (A) Representative FACS gating of hSSC lineage populations gated from singlets for living (DAPI-), non-hematopoietic (CD45<sup>+</sup>CD235a<sup>-</sup>), non-endothelial (CD31-TIE2<sup>-</sup>) cells. (B) Quantification of hSSC frequency of different fetal source tissues. (n = 1-4 donors). (C) Representative staining of CFU-F (top; Crystal violet), osteogenic (middle; Alizarin Red S) and chondrogenic (bottom; Alcian Blue) in vitro assays of prospectively isolated hSSCs from hSSCs of different fetal skeletal tissues. (D) Quantification of CFU-F assay. (n = 3-7) (E) Quantification of osteogenic assay. (n = 4-6). (F) Quantification of chondrogenic assay. (n = 2-4). (G) Representative staining of CFU-F (top; Crystal violet), osteogenic (middle; Alizarin Red S) and chondrogenic (bottom; Alcian Blue) in vitro assays of prospectively isolated skeletal lineage populations from fetal tendon tissue. (H) Representative H&E staining of a tissue section of an 8-week-old subcutaneous graft derived from tendon hSSCs able to support host-derived hematopoiesis as well as non-stem cell tendon populations forming fibrostromal tissues. (I) Dotplot of stem cell marker expression of hSSCs as defined for long bone hSSCs in this study using scRNAseq (left) and expression of markers previously used to enrich for bone stem cell-like population as found in single hSSCs from different skeletal sites (right). (J) Umap of hSSC single cell expression of Osteo./Chondro. hSSC marker SKIL and Fibro./Stromal hSSC marker HIC1 in hSSCs from different skeletal sites. (K) ScRNAseq correlation matrix of hSSCs from different fetal skeletal sites. Percentage of shared gene expression compared to growth plate SSCs shown. (L) Tracksplot of neural crest and mesoderm marker genes as found in single fetal hSSCs from frontal skull, parietal skull and suture tissue. (M) Matrixplot showing enhanced gene expression of hematopoiesis-supporting factors in stromal hSSCs compared to other subtypes. (N) Heatmap of top100 genes per Leiden cluster and their expression in hSSCs from different fetal skeletal tissues. (O) Stacked violin plots of marker genes for FSK, PSK and Suture hSSCs. (P) Tracksplot of tendon-related marker genes as found in tendon-derived single fetal hSSCs. (Q) Gene pathway enrichment analysis within Leiden clusters. (R) ForceAtlas2 clustering of fetal hSSCs sub-grouped for endochondral (left) and intramembranous (right) hSSCs showing Leiden cluster, PAGA trajectory inference and specific marker gene expression. (S) Gene expression of hSSCs from different fetal skeletal source tissues related to diseases with skeletal phenotypes shown as matrixplot. Plotted data shown as mean+standard deviation of the mean (SEM). Significance assessed by one-way ANOVA with Tukey's posthoc test.

**Figure S2: Identification and characterization of hSSC variants and lineage subtypes, Related to Figure 2.**



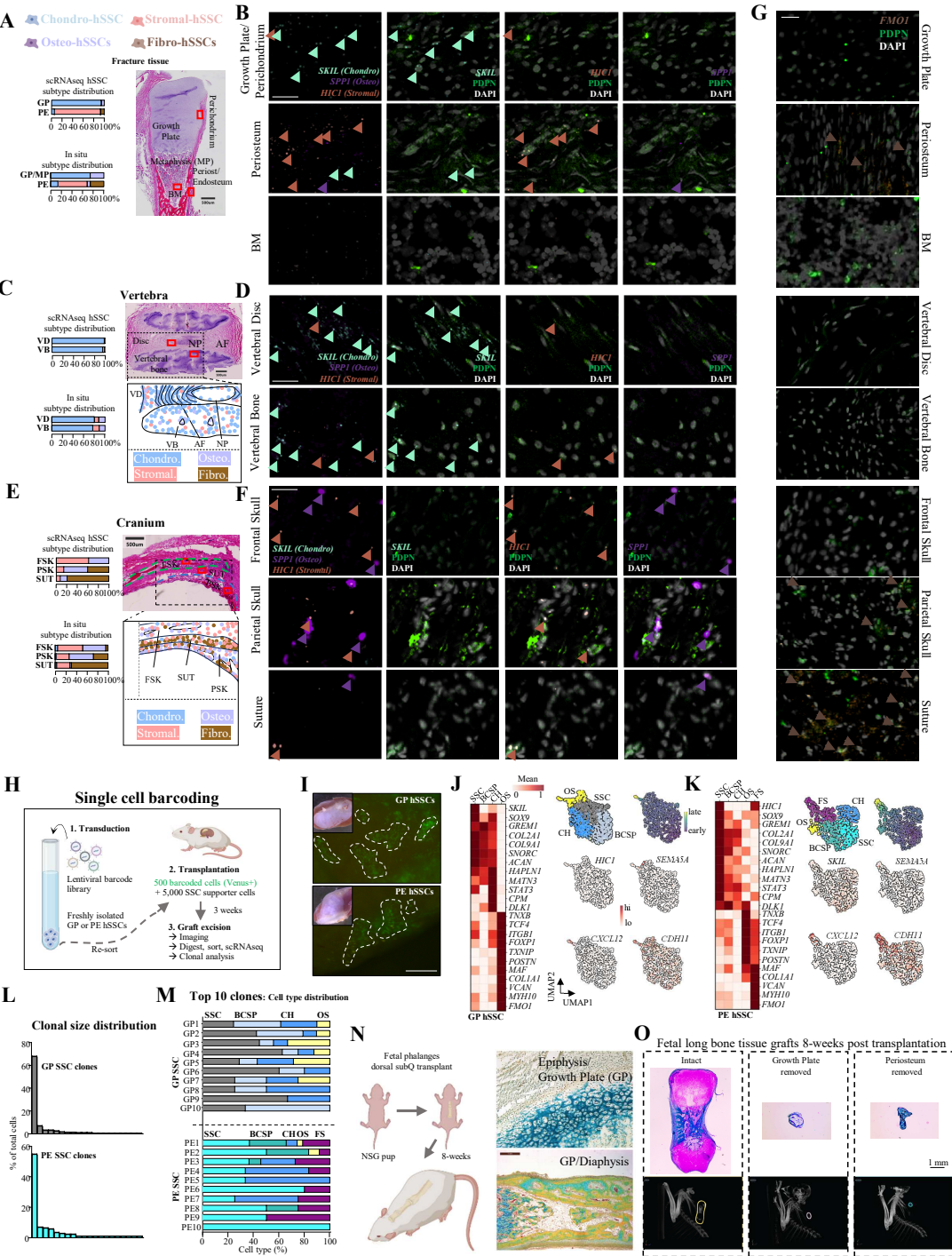
**Figure S2. Identification and characterization of hSSC variants and lineage subtypes, Related to Figure 2.** (A) Schematic of hSSC lineage tree as reported by Chan et al. Cell 2018 with cell surface marker expression of each population. (B) Previously reported skeletal stem and progenitor cell markers as expressed throughout lineage populations. (C) CytoTRACE analysis of scRNAseq data based on Leiden cluster annotated cell types and their respective inferred developmental potential. Truncated violin plots. Significance by one-way ANOVA with Tukey's posthoc test. (D) Index-sort analysis of scRNAseq data from skeletal lineage populations based on Leiden clusters. (E) Distribution of Leiden cluster populations within index-sorted prospective gates of hSSCs, hBCSPs and hOPs. (F) Five-day in vitro osteogenesis of freshly purified bone marrow stromal cells (unsorted, plastic adherent) and sorted hSSCs, OPCs (PDPN-CD146<sup>+</sup>CD73<sup>-</sup>), and SPCs (PDPN-CD146<sup>hi</sup>CD73<sup>lo</sup>) stained with Alizarin Red S. (G) Eight-week in vivo ossicles formed by OPCs (PDPN-CD146<sup>+</sup>CD73<sup>-</sup>) and SPCs (PDPN-CD146<sup>hi</sup>CD73<sup>lo</sup>) sectioned and stained for H&E. (H) Schematic of fetal growth plate with anatomical subregions and distribution of scRNAseq data based annotated cell types based on previous gene expression data of anatomically distinct GP zones. (I) GP zone-specific gene expression. (J) Proliferative gene marker expression in annotated cell types. (K) Expression of selected genes in UMAP clustering of single skeletal stem cell lineage tree populations. (L) Representative gating of hSSC subtypes within CD45-CD235a-CD31-TIE2-PDPN<sup>+</sup>CD146<sup>+</sup>-hSSCs. (M) Leiden clustering of scRNAseq data of prospectively isolated hSSC and confirmation of cluster specific markers in prospectively isolated subpopulation based on variance in CD73/CD164 cell surface marker expression. n=2 per population. Data shown as mean + SEM. (N) Flow cytometric readout of isolated hSSCs from different skeletal sites treated with short-term pro-osteogenic stimuli in vitro and distribution of hSSC subtypes. (O) Updated schematic of the human SSC lineage tree incorporating new findings. Scale bars = 50µm.

**Figure S3: Distinct features and localization of growth plate and periosteal human skeletal stem cells, Related to Figure 3.**



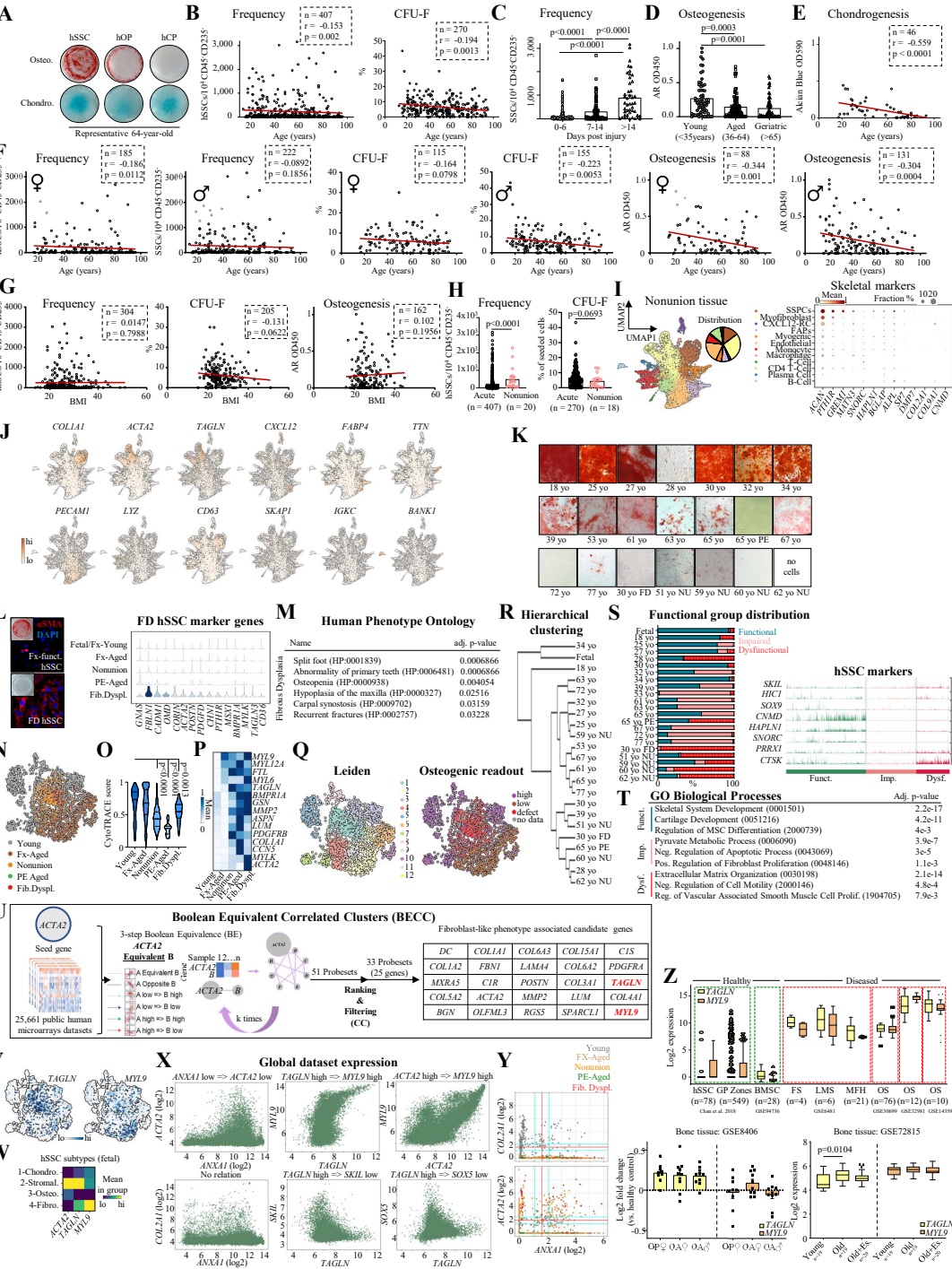
**Figure S3. Distinct features and localization of growth plate and periosteal human skeletal stem cells, Related to Figure 3.** (A) Photograph of human fetal long bone with annotated anatomical subregions (left). Representative colony-forming unit fibroblasts (Crystal violet stain; top) and osteogenic potential (Alizarin red stain; bottom) of stromal vascular fraction within defined anatomical subregions (right). (B) Representative staining of CFU-F, osteogenic and chondrogenic in vitro assays of prospectively isolated hSSCs from GP or PE fetal long bone tissue. (C) Quantification of colony-forming unit ability of prospectively isolated hSSCs, hCPs and hOPs from GP and PE fetal long bone tissue. (n = 3-7). Data shown as mean + SEM. Comparisons by two-sided Student's t-test adjusted for non-normality (Mann-Whitney test) or unequal variances (Welch's test) where appropriate. (D) Heatmap of scRNAseq data top250 genes expressed in hSSCs from GP, PE or non-dissected fetal long bone tissue. (E) Osteo./Chondro. and Fibro./Stromal hSSC marker genes and their expression in GP and PE hSSCs. (F) Expression of periosteal marker genes in scRNAseq derived UMAP of Osteo./Chondro. and Fibro./Stromal hSSCs. (G) CellChat algorithm analysis of scRNAseq data inferring ligand-receptor interactions between Osteo./Chondro. and Fibro./Stromal hSSCs from GP and PE. (H) Strength of interactions as determined by CellChat. (I) Overview of genes used to infer ligand-receptor interactions between hSSC subtypes. (J) Probability of specific ligand-receptor interactions as determined by CellChat. (K) Ligand genes clustered by signaling groups. (L) H&E stain of fetal phalange from an adjacent section used for Nanostring CosMX single cell spatial analysis. (M) UMAP plot of cell clusters and cell cluster distribution within analyzed tissues in CosMX analysis. (N) UMAP plots showing individual clusters with distribution of SOX9+PDGFR+ cells (red). (O) Expression of selected ligands in SOX9+PDGFR+ cells in subregions as well as expression of respective receptors in surrounding cells. Scale bar = 1mm.

**Figure S4: In situ identification of hSSC subtypes and in vivo clonal tracking, Related to Figure 4.**



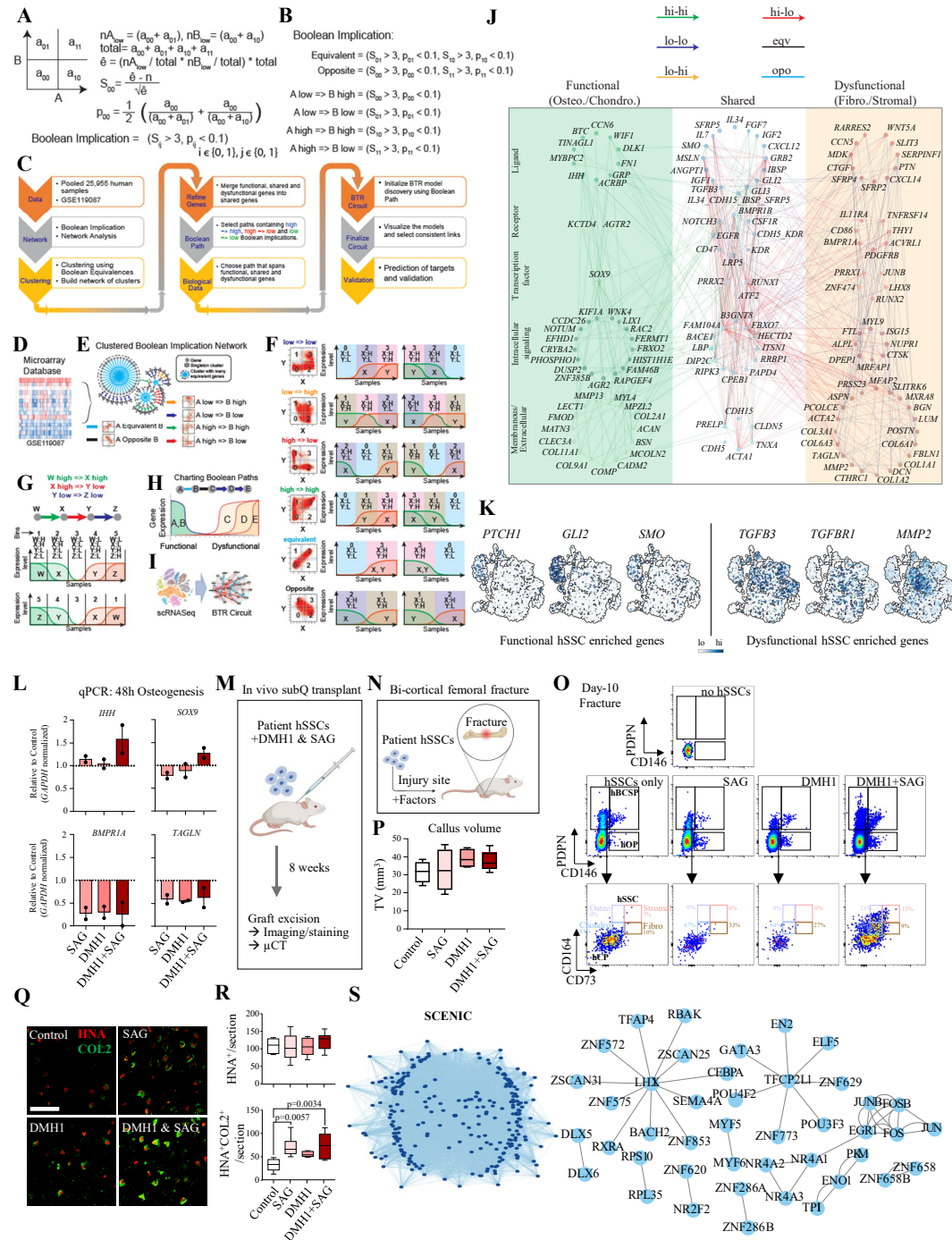
**Figure S4. In situ identification of hSSC subtypes and in vivo clonal tracking, Related to Figure 4.** (A) H&E staining of a human fetal long bone section depicting areas for in situ analysis (red squares) as well as composition of hSSC subtypes in distinct subregions as inferred by scRNAseq and in situ readout (relative abundance based on analyzed sections). (B) Representative in situ localization of chondrogenic (*SKIL*<sup>+</sup>PDPN<sup>+</sup>), osteogenic (*SPPI*<sup>+</sup>PDPN<sup>+</sup>), and stromal (*HIC1*<sup>+</sup>PDPN<sup>+</sup>) hSSC subtypes in distinct anatomical regions of long bones. (C) H&E staining of a human fetal vertebra section depicting areas for in situ analysis (red squares) as well as composition of hSSC subtypes in distinct subregions as inferred by scRNAseq and in situ readout. NP: Nucleus pulposus, AP: Annulus fibrosus. (D) In situ localization of chondrogenic (*SKIL*<sup>+</sup>PDPN<sup>+</sup>), osteogenic (*SPPI*<sup>+</sup>PDPN<sup>+</sup>), and stromal (*HIC1*<sup>+</sup>PDPN<sup>+</sup>) hSSC subtypes in distinct anatomical regions of the spine. (E) H&E staining of a human fetal cranial bone section depicting areas for in situ analysis (red squares) as well as composition of hSSC subtypes in distinct subregions as inferred by scRNAseq and in situ readout. (F) In situ localization of chondrogenic (*SKIL*<sup>+</sup>PDPN<sup>+</sup>), osteogenic (*SPPI*<sup>+</sup>PDPN<sup>+</sup>), and stromal (*HIC1*<sup>+</sup>PDPN<sup>+</sup>) hSSC subtypes in distinct anatomical regions of the cranium. (G) In situ localization of the fibrogenic (*FMOI*<sup>+</sup>PDPN<sup>+</sup>) hSSC subtype in various skeletal regions. All data is representative of tissue from two distinct donors across three sections per site. Scale bars = 25µm. (H) Schematic of experimental in vivo barcoding approach to track clonal dynamics of GP and PE hSSCs. (I) Representative images of grafts with clonal colonies of barcoded and Venus-labeled hSSCs under the renal capsule and brightfield image of graft under renal capsule (inset). Scale bar: 0.5 mm. (J) Matrixplot (left) and UMAPs (right) showing trajectory inference and expression of cell type specific marker genes for barcoded GP hSSC derived grafts. (K) Matrixplot (left) and UMAPs (right) showing trajectory inference and expression of cell type specific marker genes for barcoded PE hSSC derived grafts. (L) Clonal size distribution of all clones containing hSSCs and at least two cells with identical barcodes. GP n=46 and PE n=21. (M) Distribution of cellular subsets within top 10 most abundant clones of GP and PE hSSCs. CH: chondrogenic; OS: osteostromal; FS: fibrostromal. (N) Schematic of dorsal subcutaneous transplantation of fetal long bones elements into NSG mice (left) and representative Movat's Pentachrome staining of epiphyseal GP and diaphysis after explant of bones 8-weeks post-transplant. Scale bar: 5 mm. (O) Subcutaneous fetal long bone grafts 8-weeks post transplantation either intact, or with prior removal of growth plate or periosteum. Top images show tissue sections with overview stain. Bottom images are micro-CT images of graft tissue in vivo.

**Figure S5: Aging manifests a fibrogenic lineage fate in human skeletal stem cells, Related to Figure 5.**



**Figure S5. Aging manifests a fibrogenic lineage fate in human skeletal stem cells, Related to Figure 5.** (A) In vitro multipotent differentiation capacity of prospectively isolated cell populations from aged patients. (B) Left: Association of abundance of freshly isolated patient hSSCs and age of donor based on flow cytometric analysis of freshly processed callus tissue (n = 407). Right: Association of CFU-F and patient age in primary hSSCs (n=270). All correlations assessed by Spearman test. (C) Frequency of hSSC abundance in fracture callus tissue based on FACS analysis. (0-6 days post injury, n=262; 7-14 days, n=93; >14 days, n=46) (D) In vitro mineralization capacity of hSSCs grouped by patient age (young: <35 years, n=65; aged: 35-65 years, n=93; geriatric: >65 years, n=62). Significances assessed by one-way ANOVA with Tukey post-hoc test. (E) Association of chondrogenic in vitro potential and age of hSSC donor. (F) Correlation of hSSC frequency, CFU-F capacity and in vitro osteogenic capacity with age of patient separated into female and male donors. (G) Correlation of BMI with hSSC frequency, CFU-F capacity and in vitro osteogenic potential. All correlations assessed by Spearman test. (H) Left: Frequency of phenotypic hSSCs based on flow cytometric analysis in acute fracture callus hSSCs (n=407) and nonunion hSSCs (n=20). Right: In vitro CFU-F potential of acute fracture (n=270) versus nonunion (n=18) derived hSSCs. Data shown as mean + standard deviation of the mean (SEM). Significance assessed by Mann-Whitney test. (I) UMAP plot of annotated cells clustered by Leiden found in single cell RNA-sequenced (10X Chromium) nonunion tissue. Pie chart shows relative contribution of each cell populations to all cells captured. SSPCs: Skeletal Stem and Progenitor Cells; FAPs: Fibro-Adipogenic Progenitors; CXCL12-RCs: CXCL12-expressing Reticular Cells. Right: Dotplot of gene expression of osteochondrogenic skeletal stem and differentiation markers in each Leiden cell population. (J) UMAP plot showing Leiden cluster specific gene expression in nonunion tissue. (K) In vitro osteogenic readout of hSSCs also processed for SmartSeq2 single cell RNA-sequencing. (L) Left: Representative immunohistochemical staining of Smooth muscle actin (aSMA) in freshly cultured hSSCs from a healthy fracture (left) and Fibrous Dysplasia (right) tissue. Insets show in vitro osteogenic readouts of the same cells. Right: Genes overexpressed in hSSCs in Fibrous Dysplasia compared to hSSCs from other source tissues. (M) Gene ontology of phenotypes related to gene expression unique to Fibrous Dysplasia hSSCs. (N) Leiden clustering of single hSSCs colored by patient group. (O) CytoTRACE analysis of single cell data inferring developmental potential of hSSCs grouped by distinct source tissues. Data shown as violin plots. Statistical testing against 'Young' done by One-way ANOVA with Bonferroni posthoc test. (P) Matrixplot of gene expression related to fibrogenic phenotypes in different hSSC groups. (Q) Left: UMAP of Leiden clustering of single hSSCs. Right: UMAP colored by osteogenic readout. (R) Hierarchical clustering of donor-derived hSSCs. (S) Abundance of donor hSSCs within functional group clusters. Right: Tracksplot of hSSC related markers as found in fetal long bone hSSCs and their expression in functional subgroups of human adult fracture hSSCs. (T) Selected GO Biological Processes in top250 genes of hSSCs of different functional states based on single cell RNA-sequencing hSSC gene expression data. (U) Schematic overview of Boolean Equivalent Correlated Clusters (BECC) algorithm approach performed on the global human dataset (GSE119087). Based on a seed gene (*ACTA2*) the first step performs Boolean analysis by following the Boolean equivalent relationship three times. In the second step a score is computed for each gene which are then ranked. Finally, a threshold is imposed using the StepMiner algorithm to define a list of high-confidence genes with potentially superior performance than the seed gene (i.e., marking a fibrogenic/dysfunctional state). (V) UMAP plot of patient hSSCs with expression of *TAGLN* and *MYL9*. (W) Gene expression in fetal hSSC subtypes. (X) Boolean relationships of selected genes based on the global microarray dataset. (Y) Boolean relationships of *TAGLN* and *MYL9* with bone marker genes in hSSCs from nonunion fracture tissue, Fibrous Dysplasia and fetal bones as well as periosteal tissue. (Z) Top: Gene expression of *TAGLN* and *MYL9* in selected published datasets of healthy and pathological cells/tissues (GP Zones: Growth Plate Zones; BMSC: Bone Marrow Stromal Cell; FS: Fibrosarcoma; LMS: Leiomyosarcoma; MFH: Malignant Fibrous Histiocytoma; OS: Osteosarcoma). Bottom left: Gene expression of *TAGLN* and *MYL9* in publicly available GSE8406 dataset containing femoral trabecular bone extracted from healthy as well as osteoporotic (OP) and osteoarthritic (OA) patients. Results are shown as log2 fold change of aged-matched healthy controls versus diseased OP (n=9) or OA tissue (n=20). Dataset reports overall low fold changes with almost all data points between -0.5 and 0.5, potentially reflecting the technical limitation in early microarray technology. Genes could be confirmed by qPCR and showed higher fold changes. Bottom right: Gene expression of *TAGLN* and *MYL9* in GSE72815 dataset containing iliac crest bone biopsies (n=58) from healthy young ( $30.3 \pm 5.4$  years) and old patients ( $73.1 \pm 6.6$  years) as well as old patients that underwent three weeks of estrogen (Es.) therapy ( $70.5 \pm 5.2$  years). Data shown as whisker boxplots with Tukey distribution. Significance assessed by one-way ANOVA with Tukey's post-hoc test.

**Figure S6: Boolean based analysis of gene regulatory networks in single cells, Related to Figure 6.**



**Figure S6. Boolean based analysis of gene regulatory networks in single cells, Related to Figure 6. (A)** BooleanNet statistic is used to evaluate Boolean implication relationships between two genes, A and B.  $a_{ij}$  is the number of samples in the respective quadrants.  $nA_{low}$  and  $nB_{low}$  are number of samples where A and B is low respectively.  $S_{00}$  = BooleanNet statistic and  $p_{00}$  = error rate to test sparsity for the bottom left quadrant.  $S > 3$  and  $p < 0.1$  is used to test whether each quadrant is sparse. **(B)** Deriving six possible Boolean implication relationships using BooleanNet statistics. **(C)** Schematic pipelines of Boolean analysis performed in this manuscript. **(D)** Boolean Implication analysis was performed on the global human dataset (GSE119087) which was built from 25,955 human microarrays from diverse tissues (bulk samples). **(E)** Clustered Boolean Implication Network and six possible Boolean Implication Relationships (BIRs) between genes/clusters. **(F)** Visualization of Boolean Implication relationships by one-dimensional plots of gene expression levels assuming gene expression value change (up or down) only once along a path. Samples are ordered in two possible ways to show the relationship. X low  $\Rightarrow$  Y low can be represented in a single dimension in two possible ways: (1) X turns on first and then Y turns on along a hypothetical biological path defined by the sample order. The path begins with X low and Y low, then it transitions to X high and Y low, and finally it transitions to X high and Y high. (2) Y turns off first and then X turns off along a hypothetical biological path defined by the sample order. The path begins with X high and Y high, then it transitions to X high and Y low; finally, it transitions to both low. Similarly, the other Boolean relationships follow two possible path scenarios based on logical conclusions. **(G)** Visualization of a complex Boolean path using one-dimensional plot. (1) Three Boolean relationships between W, X, Y, and Z. (2) Graphic representation of the Boolean relationships. Edges are colored differently according to the type of Boolean relationship. (3) W turns off first and then X turns off along a hypothetical biological path defined by the sample order followed by Y turning on and Z turning on. The path begins with W high, X high, Y low and Z low. (4) The path begins with W low, X low, Y high and Z high. Z turns off first and then Y turns off along a hypothetical biological path defined by the sample order followed by X turning on and Y turning on. **(H)** Similar to panel E, four Boolean implication relationships A equivalent B, B opposite C, C low  $\Rightarrow$  D low, D low  $\Rightarrow$  E low constitute a Boolean path that can be used to develop a computational model of biological process from functional to dysfunctional. The computational model predicts how these five genes might be changing along a biological path as shown in the plot. **(I)** In the final step, selected paths from the Clustered BIN is used with BoolTraineR (BTR) which incorporates the single-cell RNAseq dataset as an input, the Boolean value of genes in the beginning of the selected pathway in MiDRcG algorithm as an initial state, and the BIRs between genes as an initial Boolean model. A gene regulatory network constructed underlying BTR for single-cell RNAseq dataset is generated. **(J)** Complex Boolean implication network (BIN) derived from large database (GSE119087) of publicly available human microarrays of human tissues for genes expressed in functional and dysfunctional hSSCs. Expression relationships: hi-hi = A high  $\Rightarrow$  B high; hi-lo = A high  $\Rightarrow$  B low; lo-hi: A low  $\Rightarrow$  B high; lo-lo: A low  $\Rightarrow$  B low; eqv. = equivalent; opo. = opposite. **(K)** Expression of Hedgehog pathway specific genes and TGF $\beta$  pathway specific genes in single cell RNA-sequencing data of functional and dysfunctional hSSCs. **(L)** Quantitative PCR of primary hSSCs treated with factors or control for 48h in vitro. n=2 donors. **(M)** Schematic of in vivo subcutaneous grafting of patient hSSCs. **(N)** Schematic of in vivo fracture site grafting of patient hSSCs. **(O)** Flow cytometric readout of digested callus tissue using the hSSC lineage panel. **(P)** Micro-CT based callus volume quantification at day-14 after fracture. **(Q)** Immunohistochemistry identifying hSSC-derived cells stained for HNA (Human Nuclear antigen, red) and Collagen-2 (COL2, green) in fracture callus tissue of different experimental groups. **(R)** Quantification of HNA and COL2 double-positive cells across calluses. One-way ANOVA with Dunnett's posthoc test. **(S)** Prediction of enriched transcription factor networks using SCENIC based on single cell RNA-sequencing data of patient hSSCs for comparison to Boolean method. Left: whole network. Right: selected top 50 important transcription factors and their interactions. Scale bar = 50 $\mu$ m.